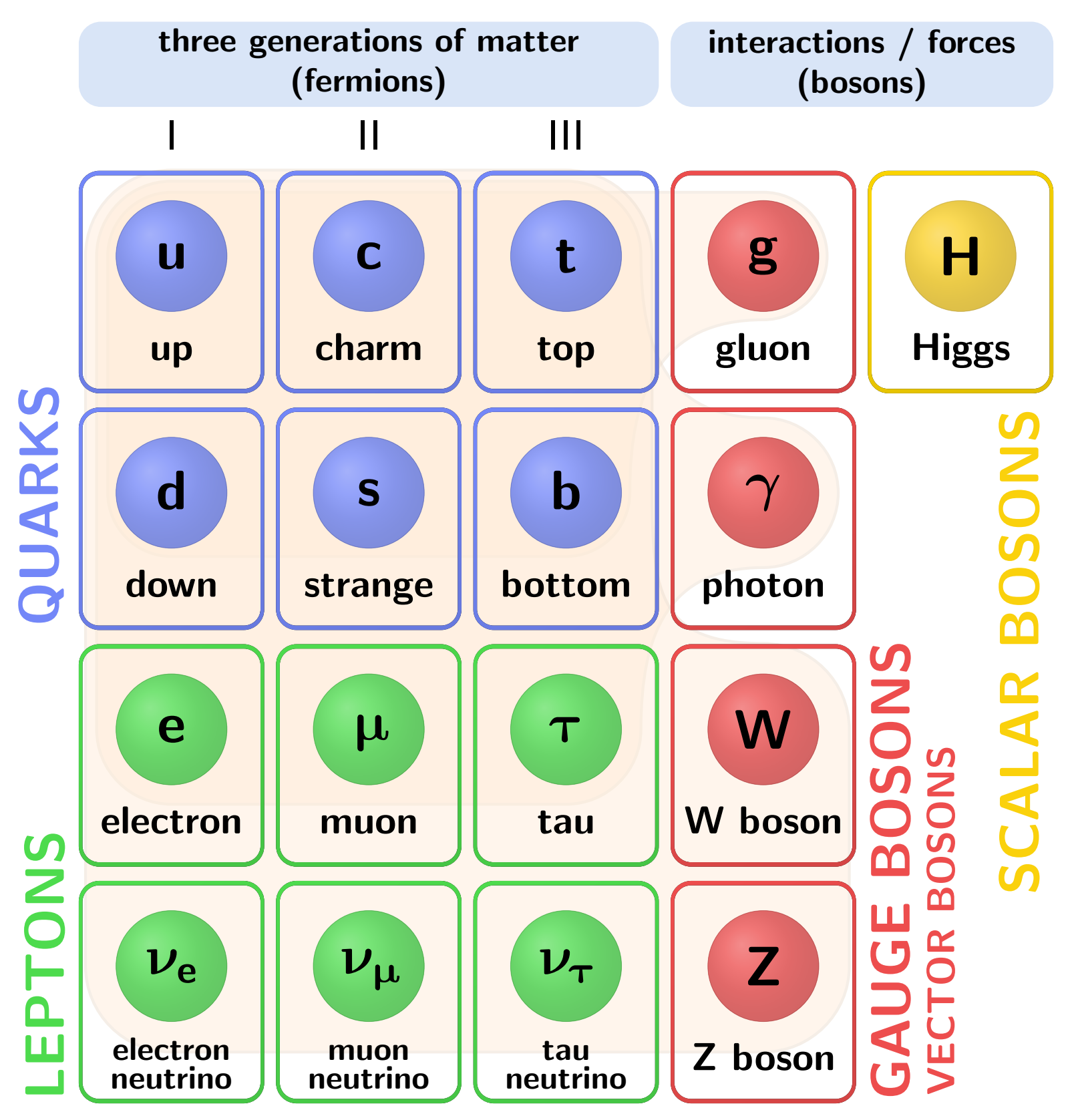
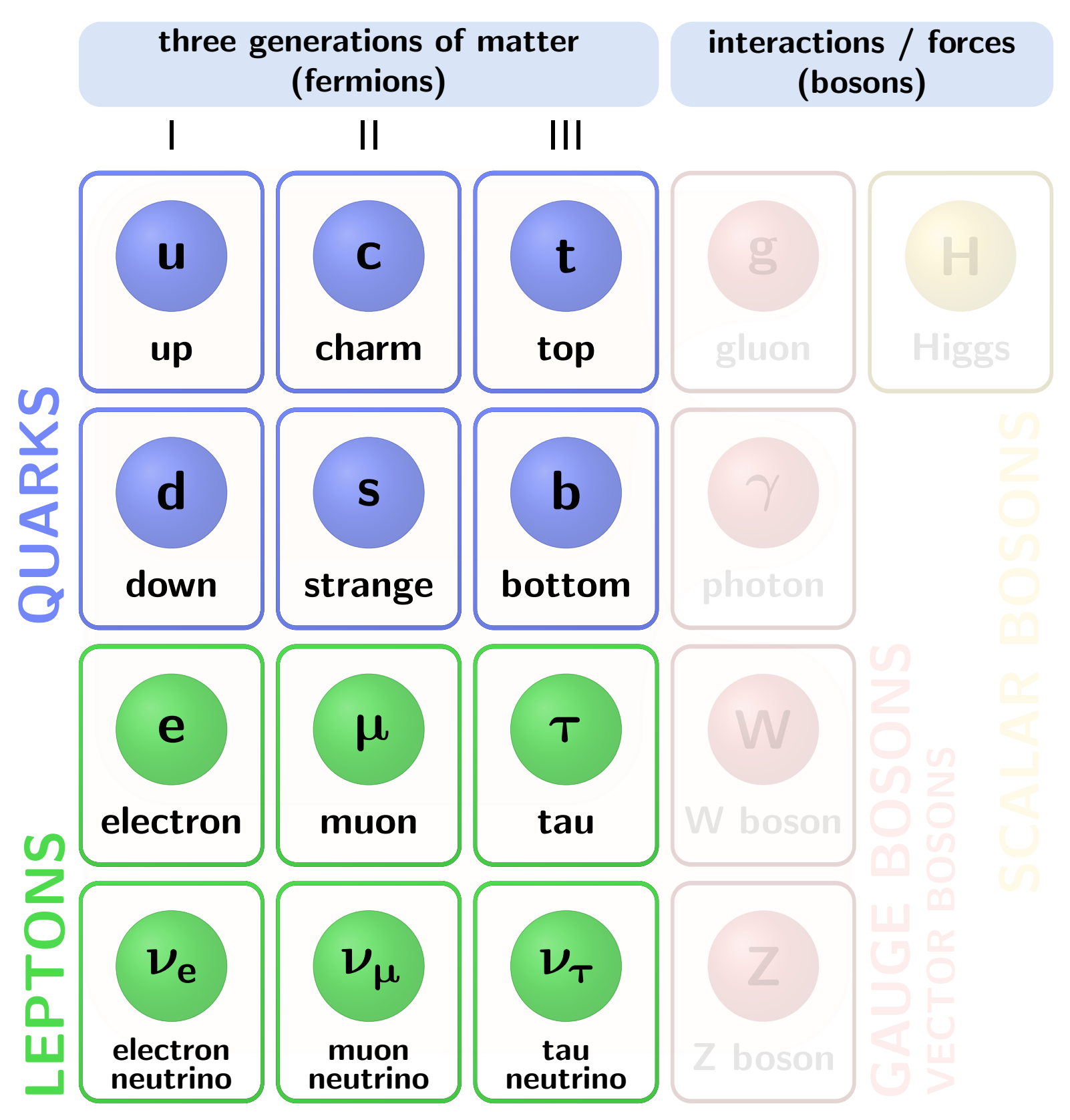
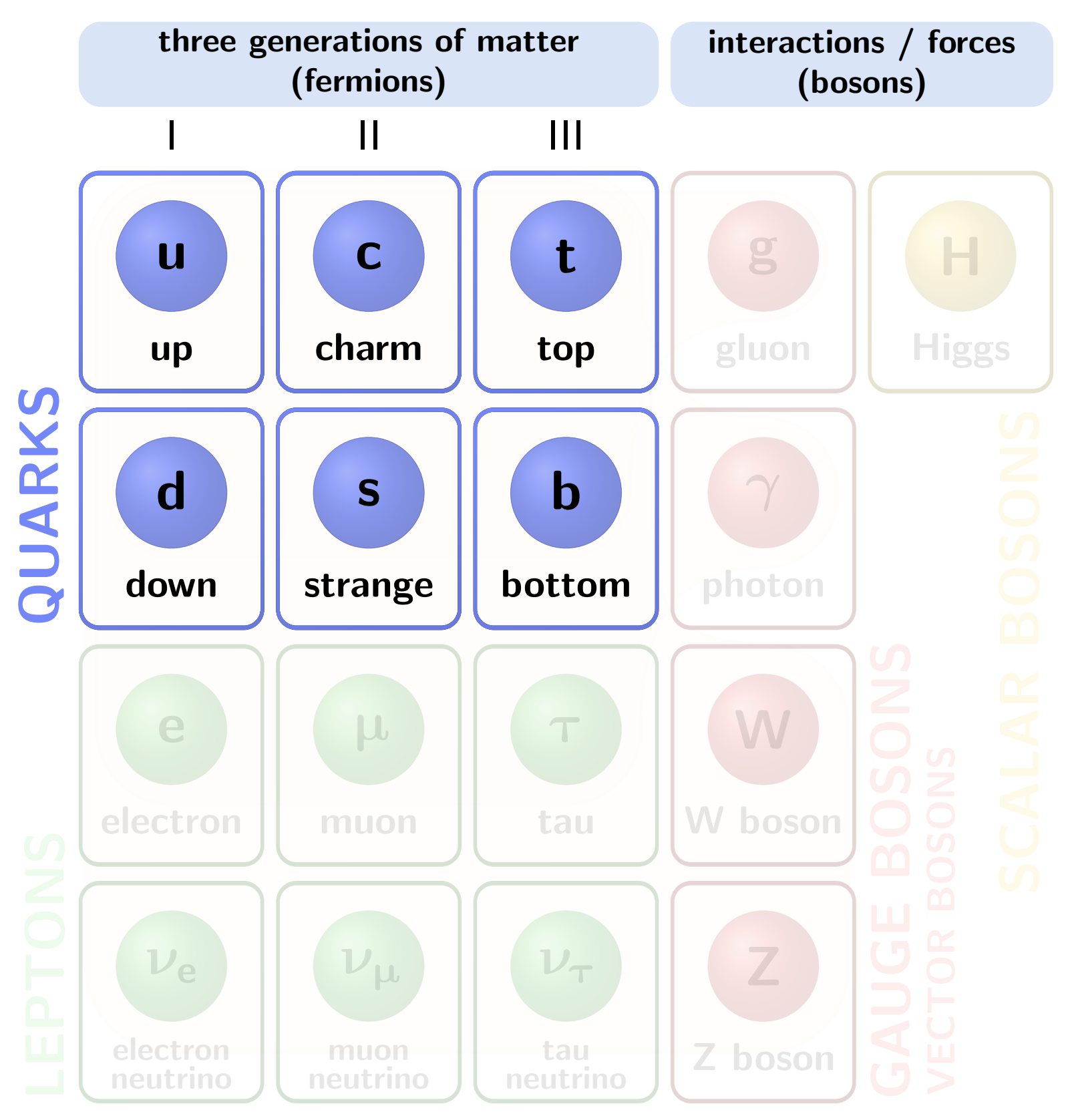
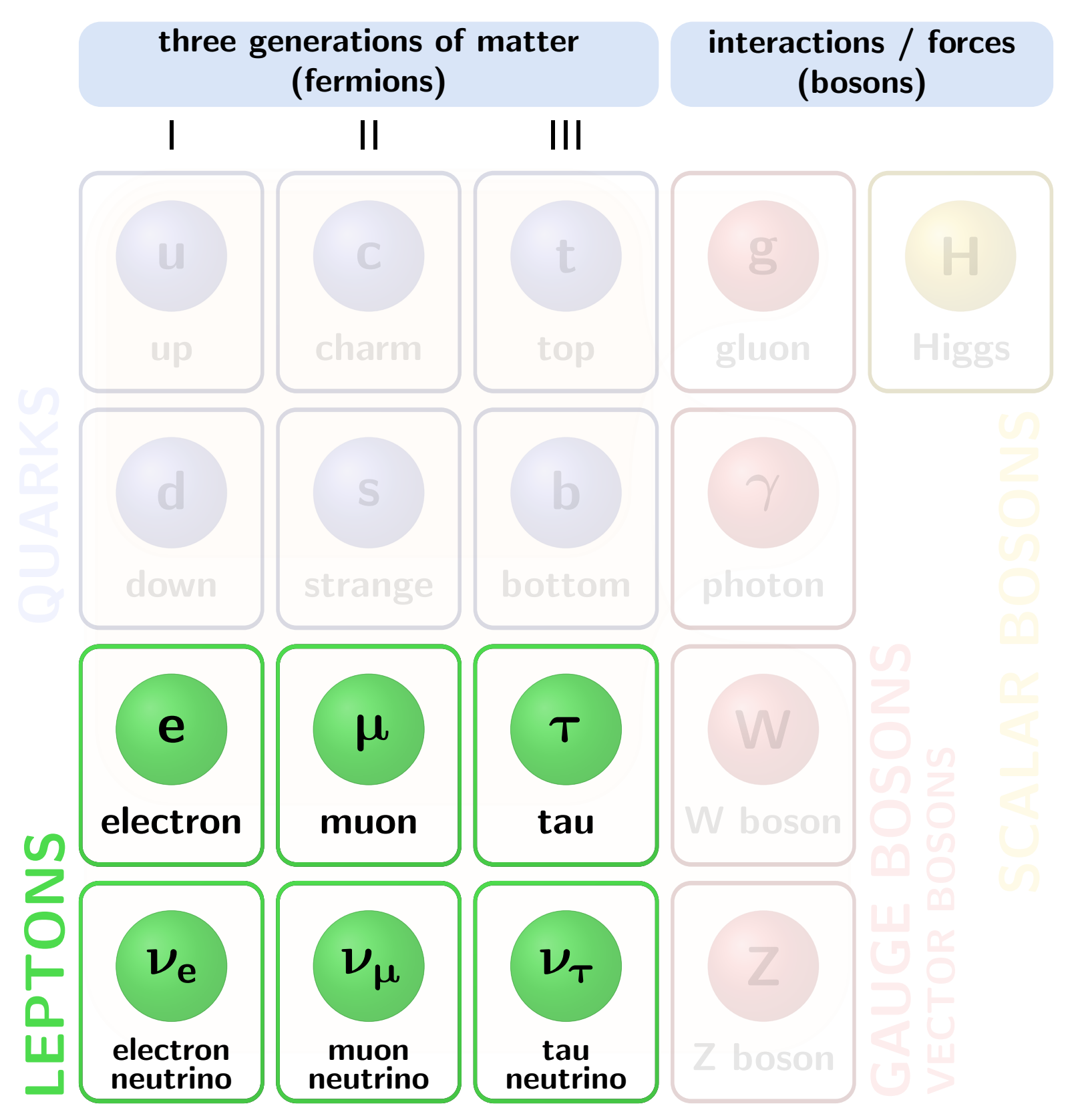
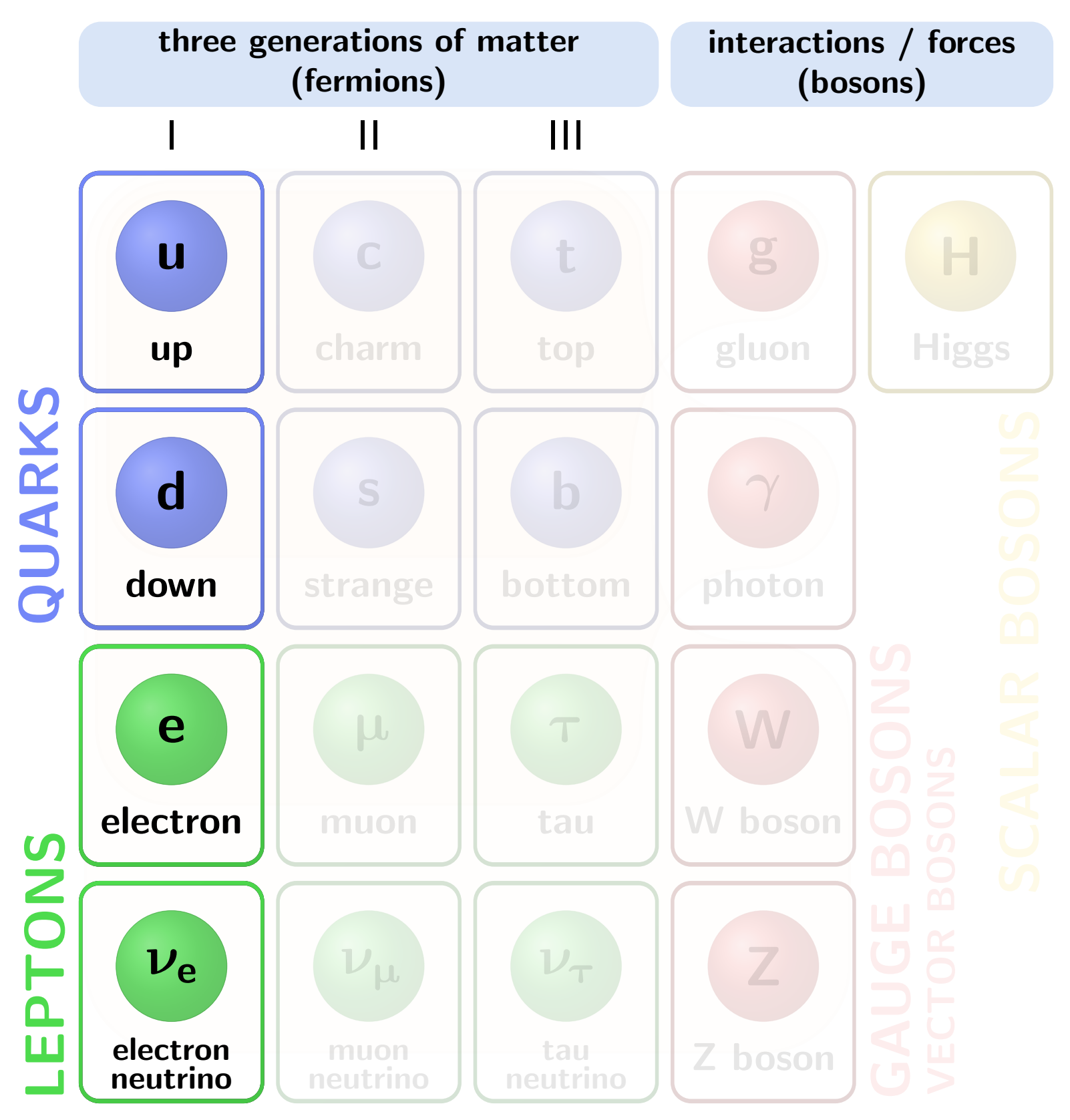
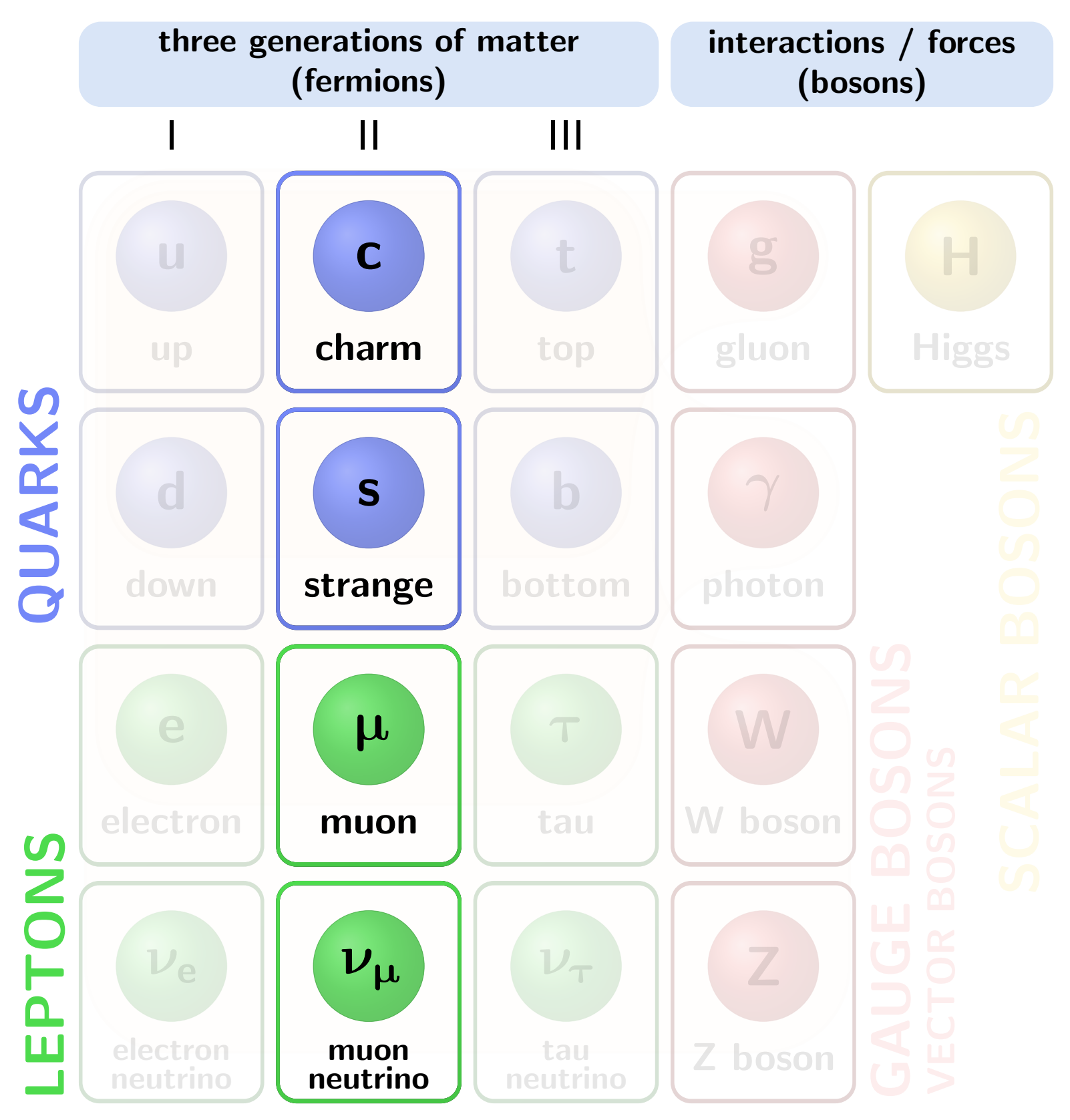

three generations of matter
(fermions)
interactions / forces
(bosons)
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VECTOR BOSONS
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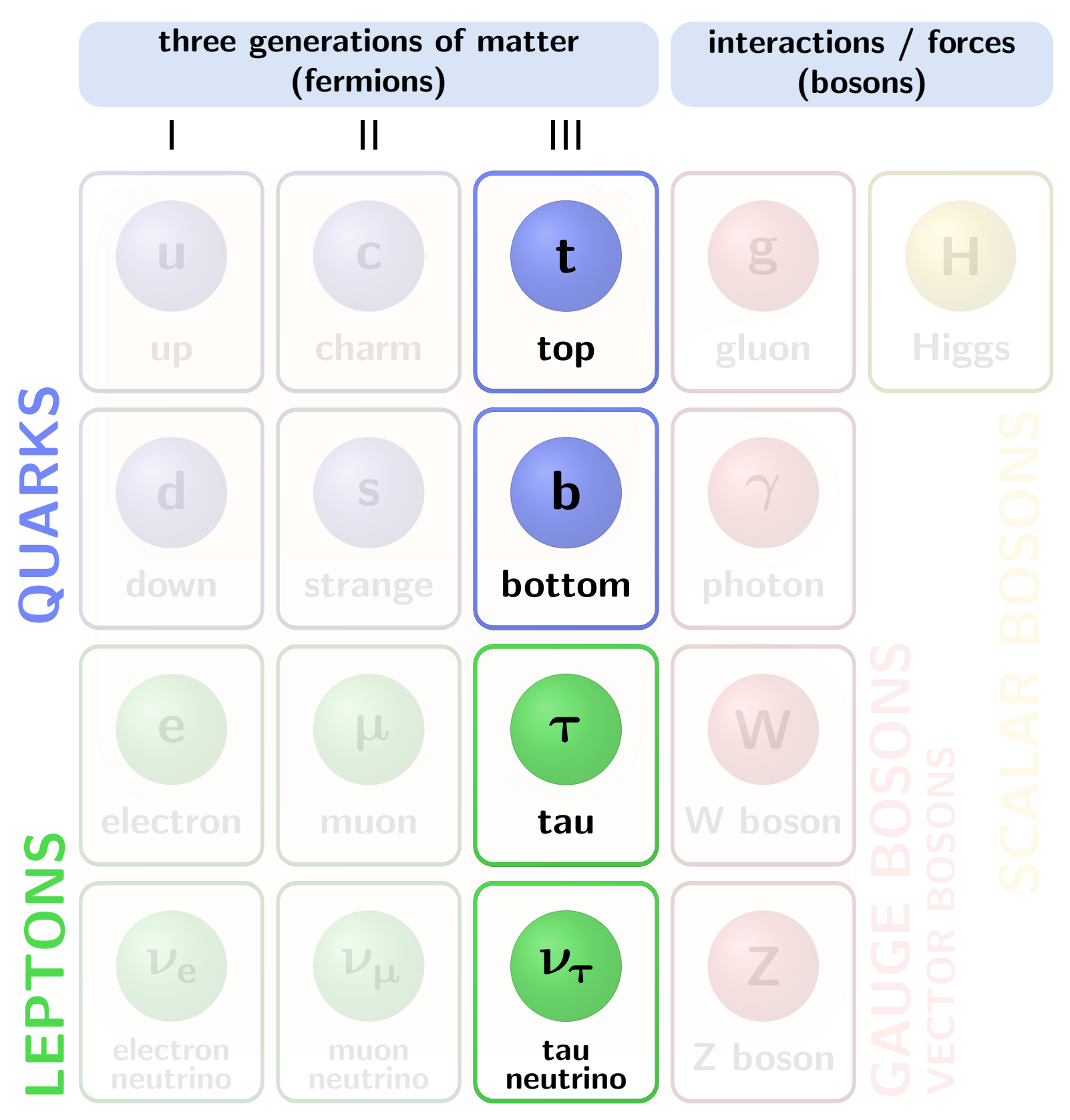

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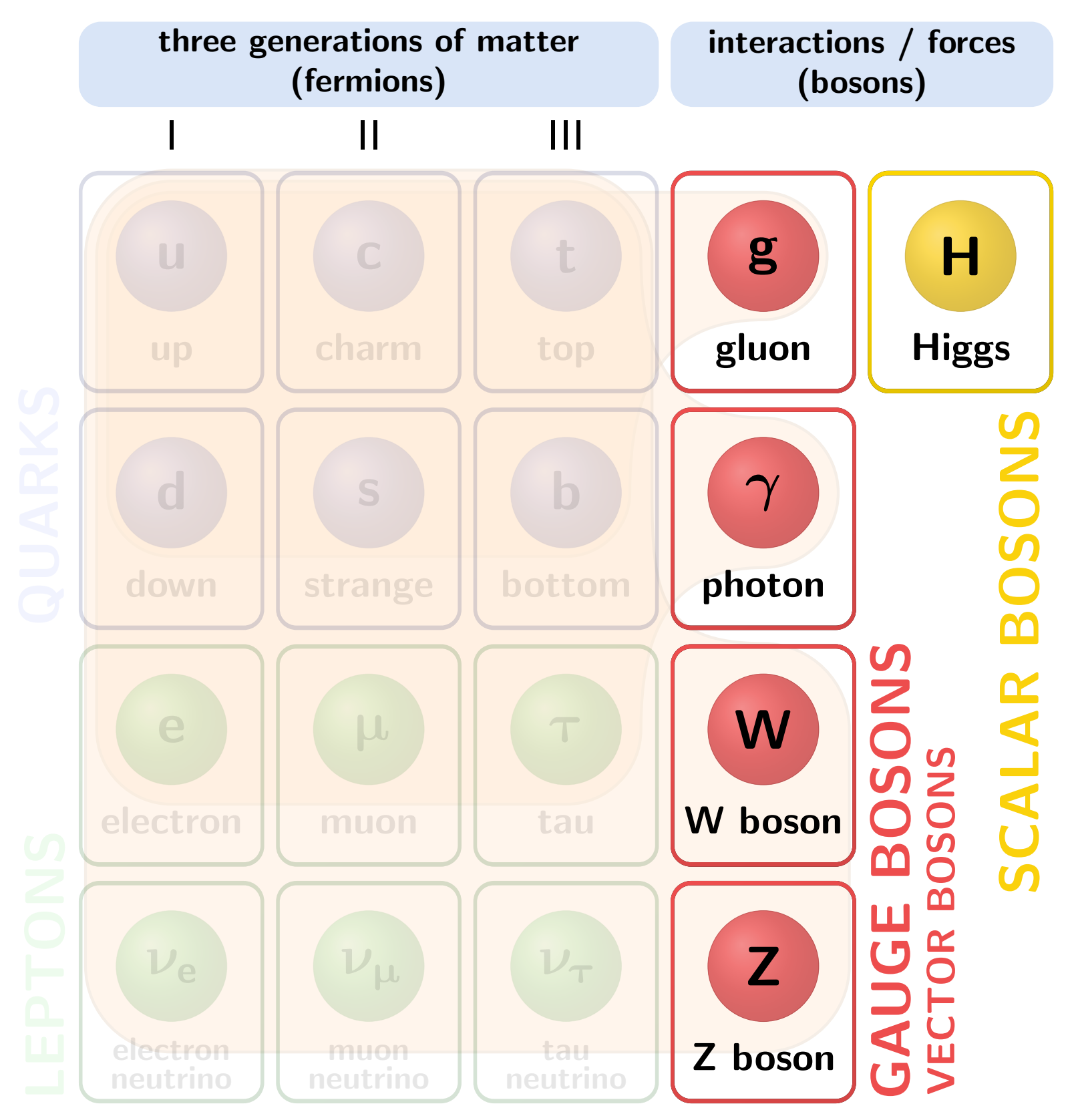

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interactions / forces (bosons)
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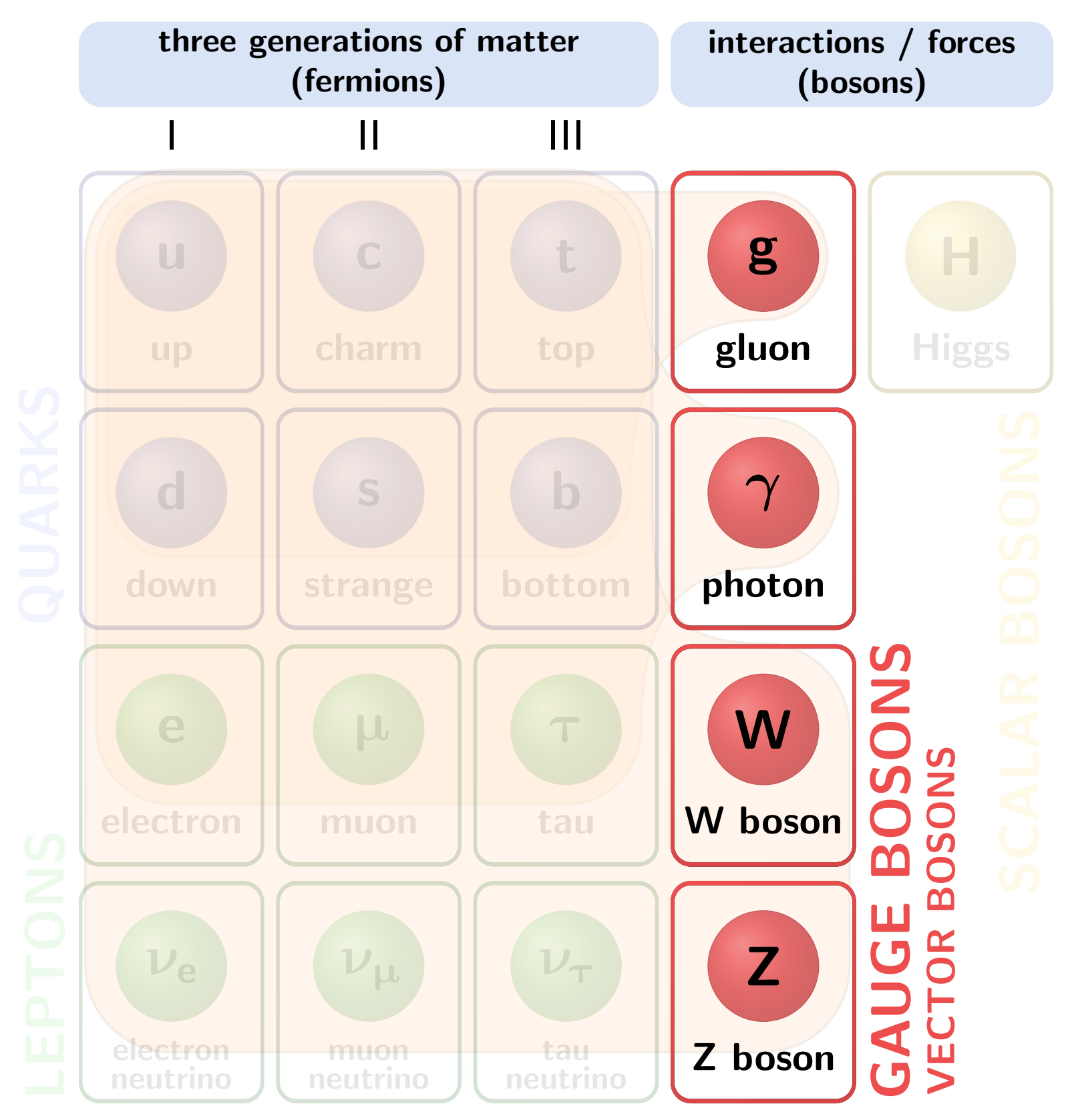

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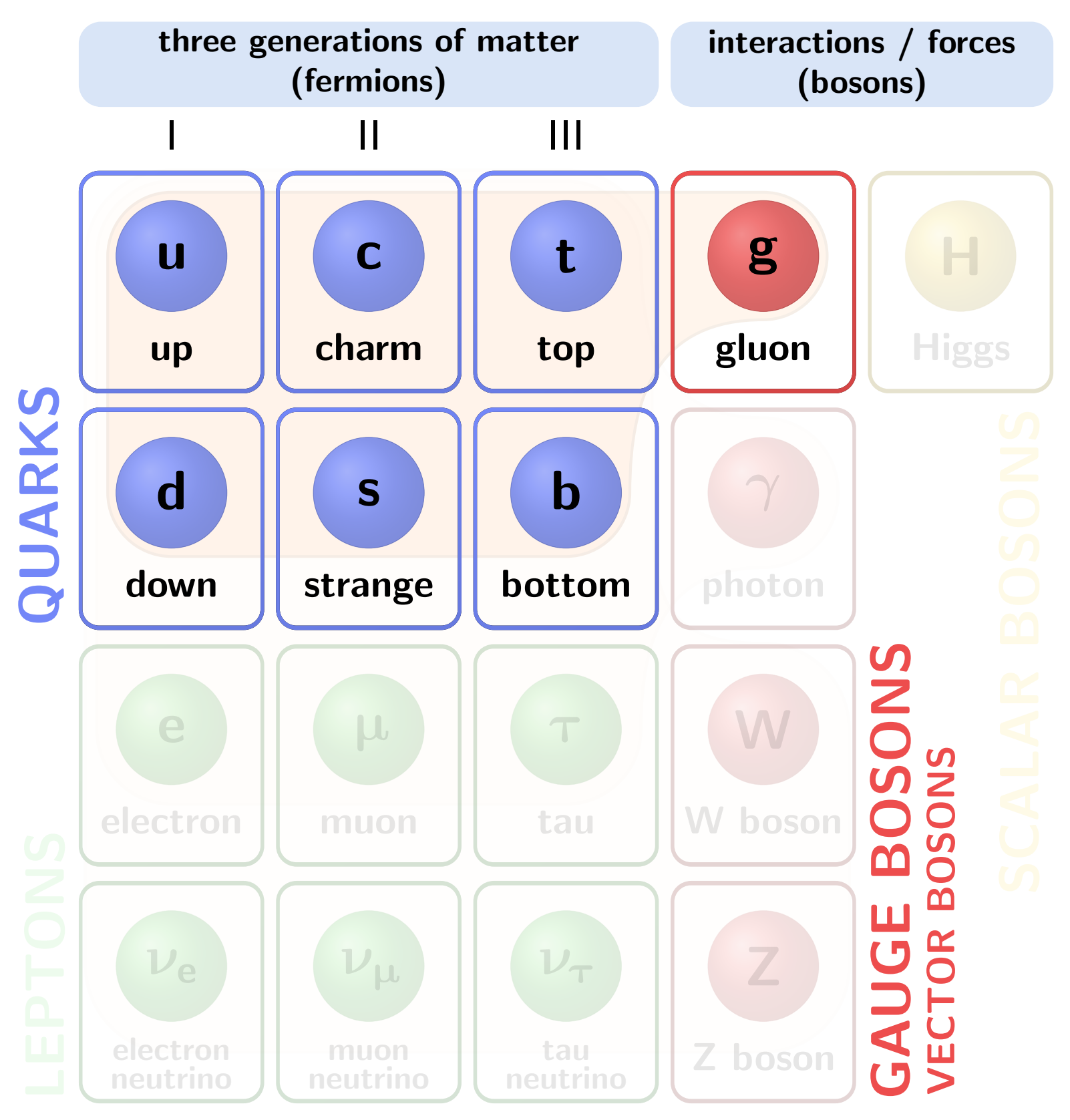

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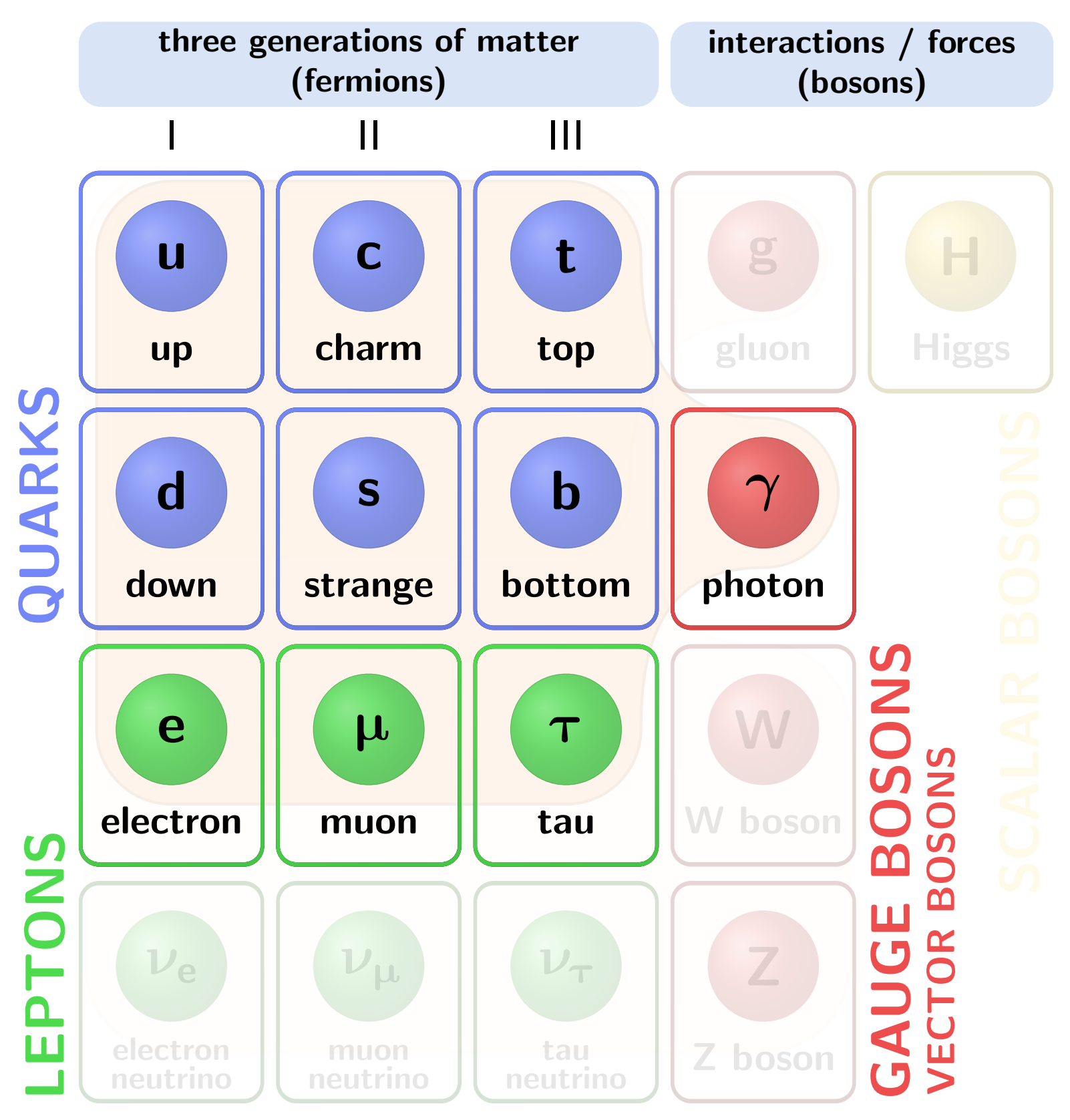

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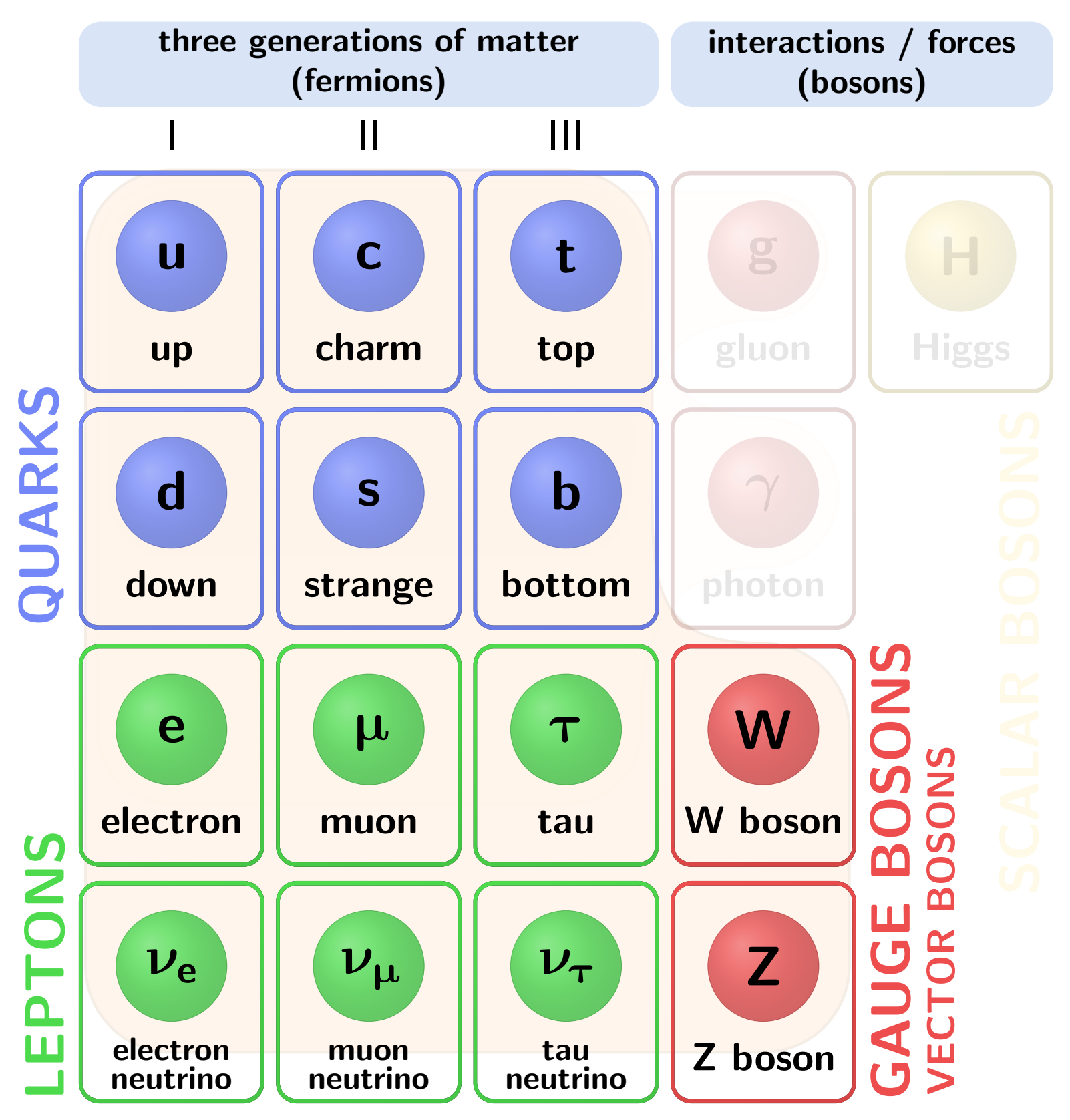

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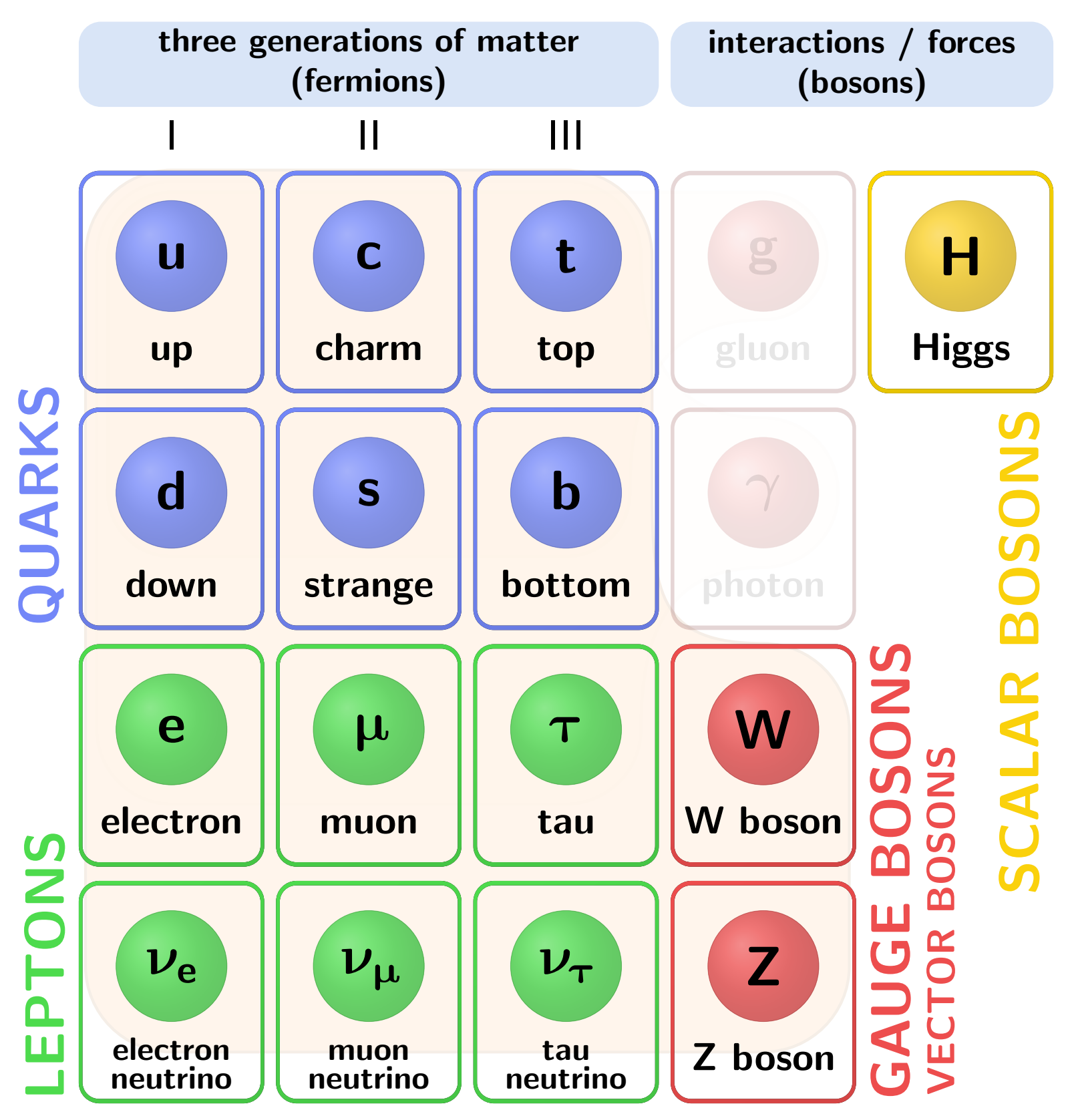

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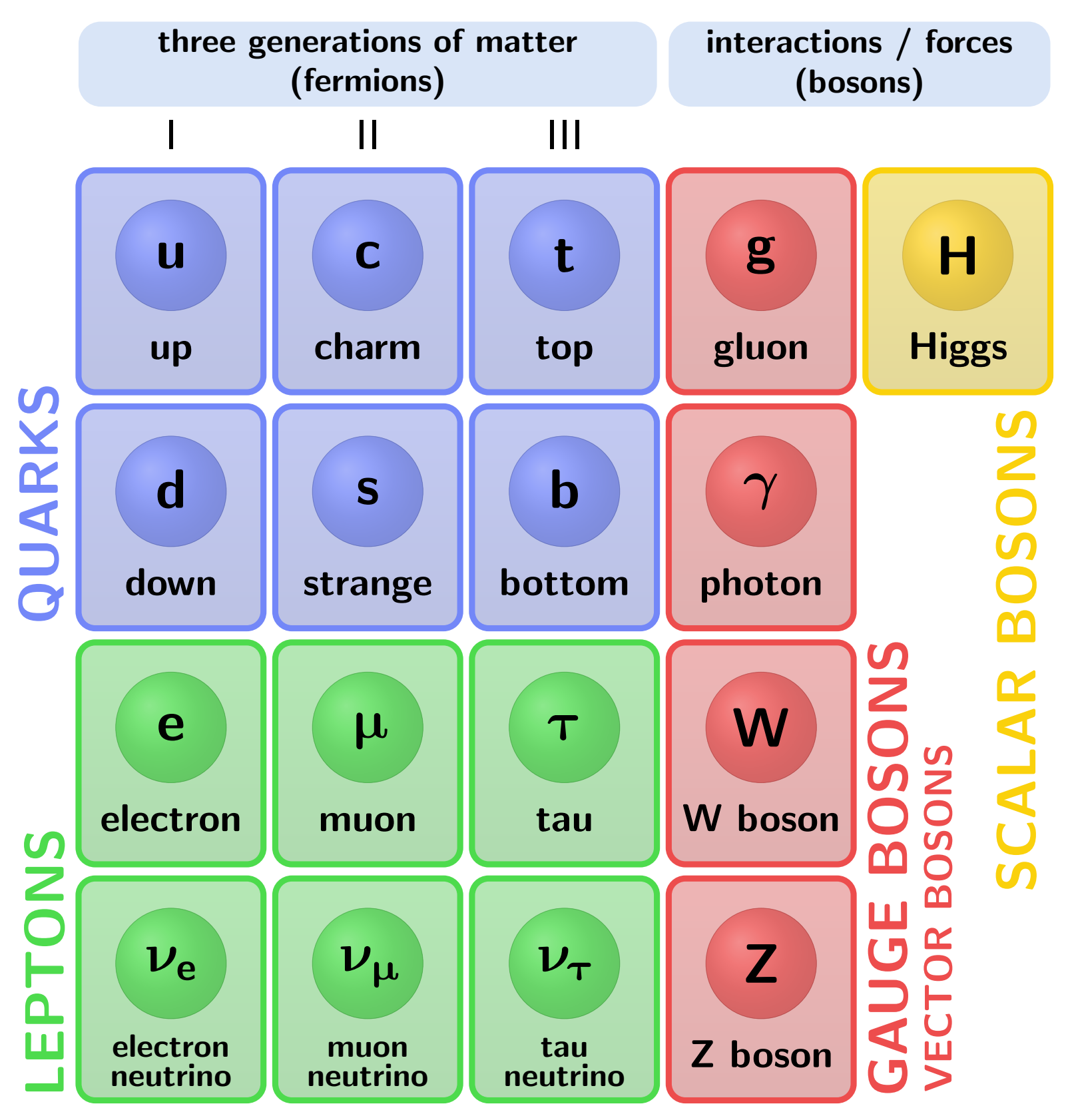

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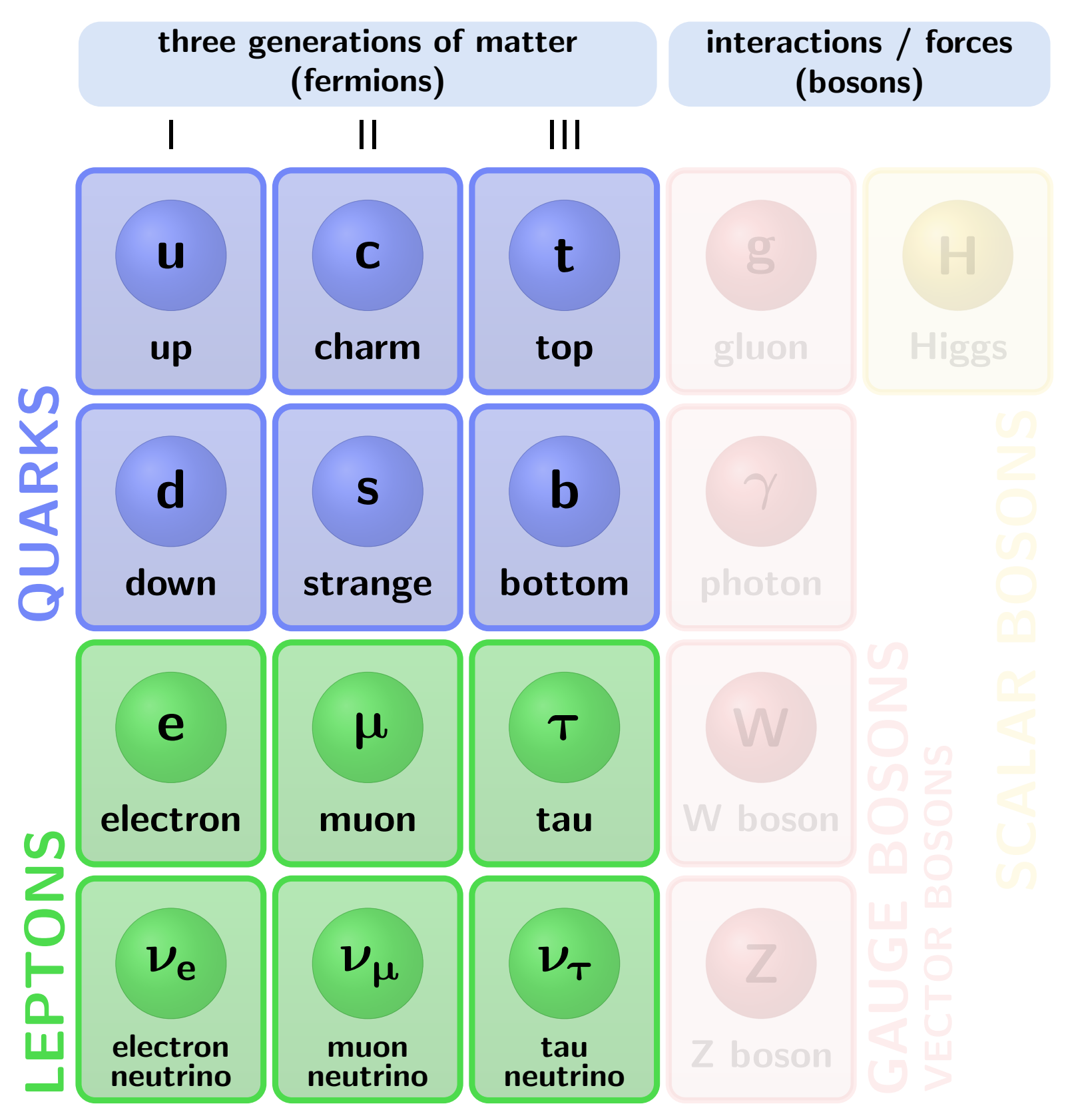

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SCALAR BOSONS

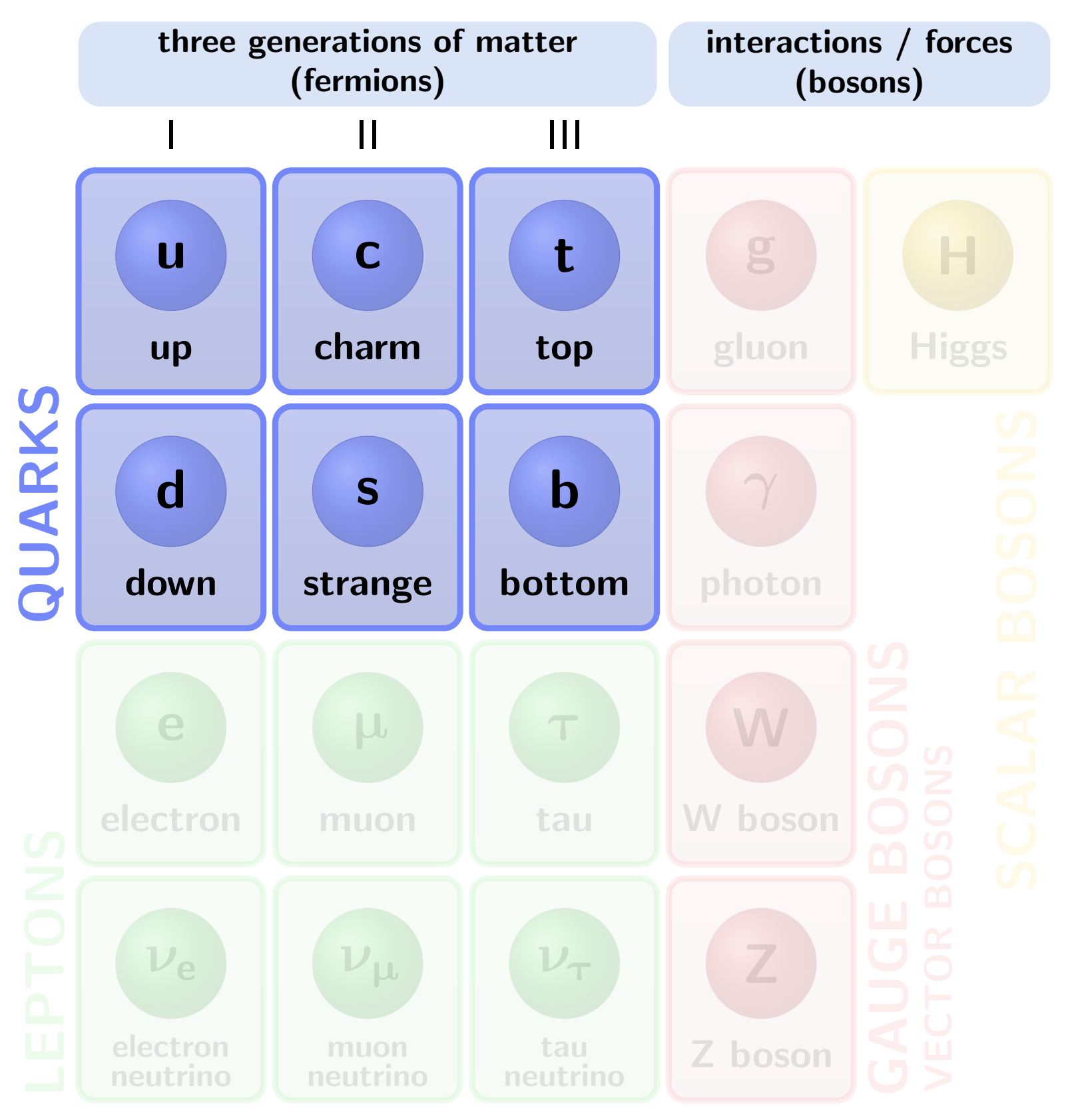

three generations of matter
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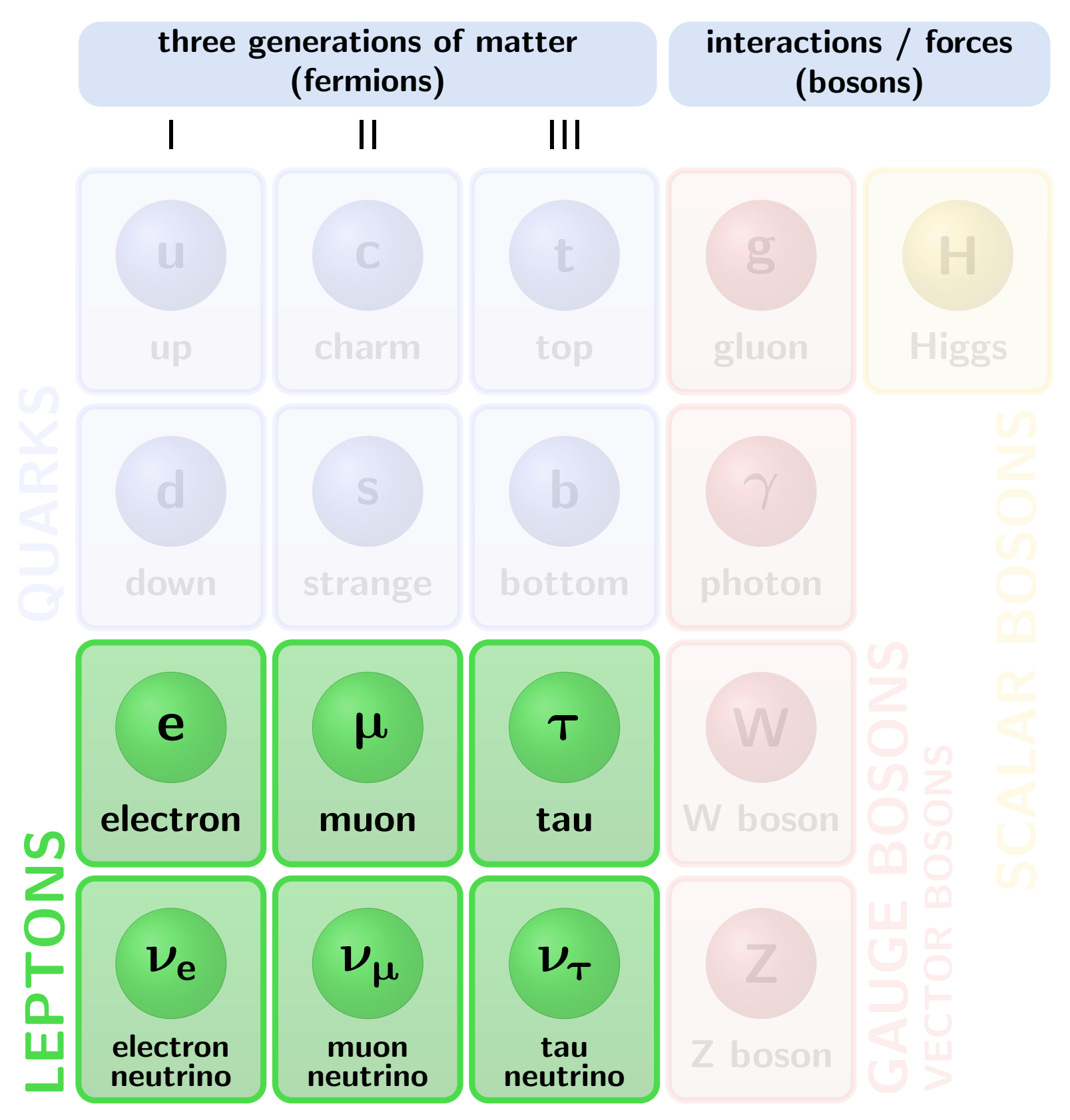

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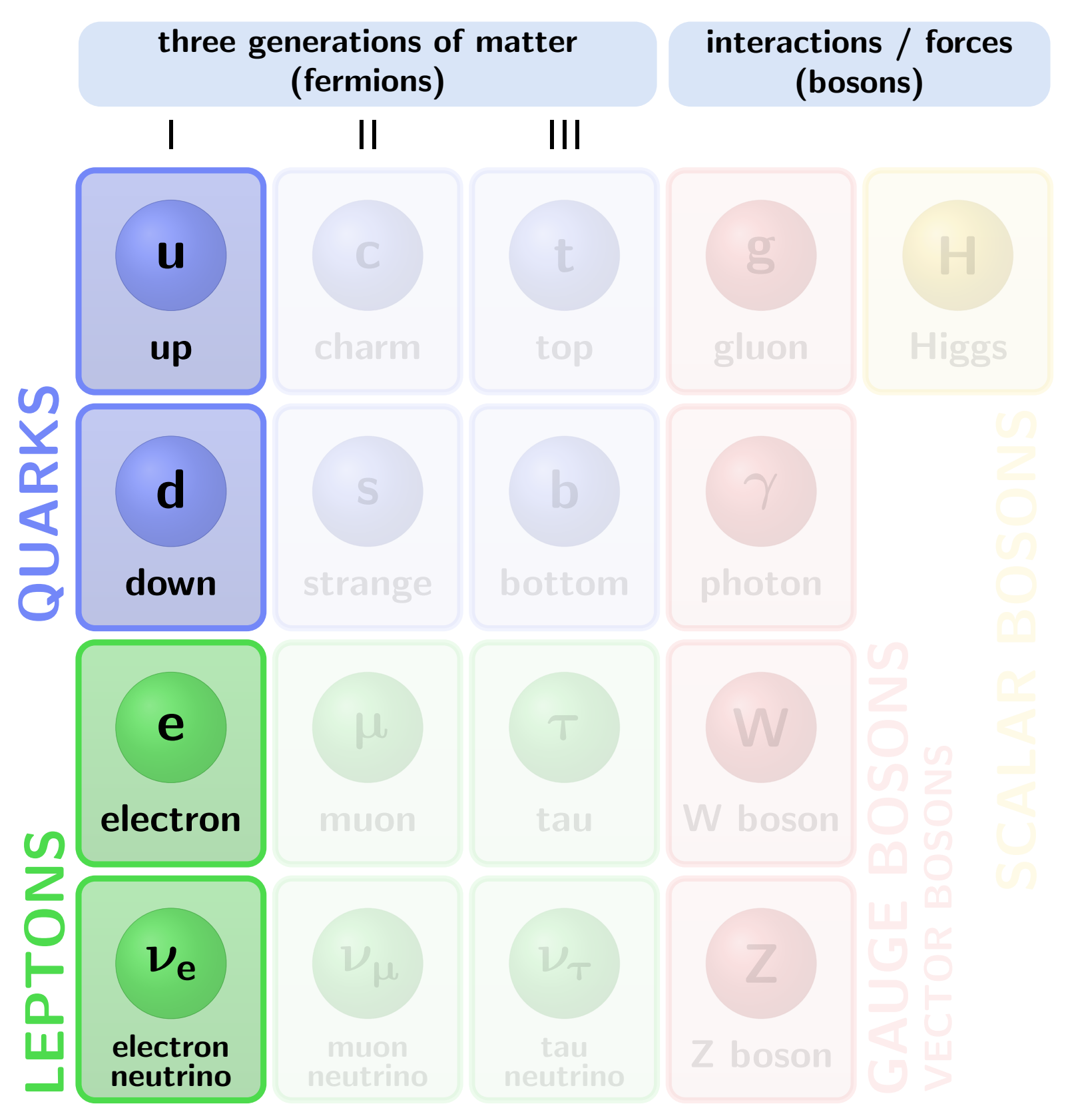

three generations of matter (fermions)
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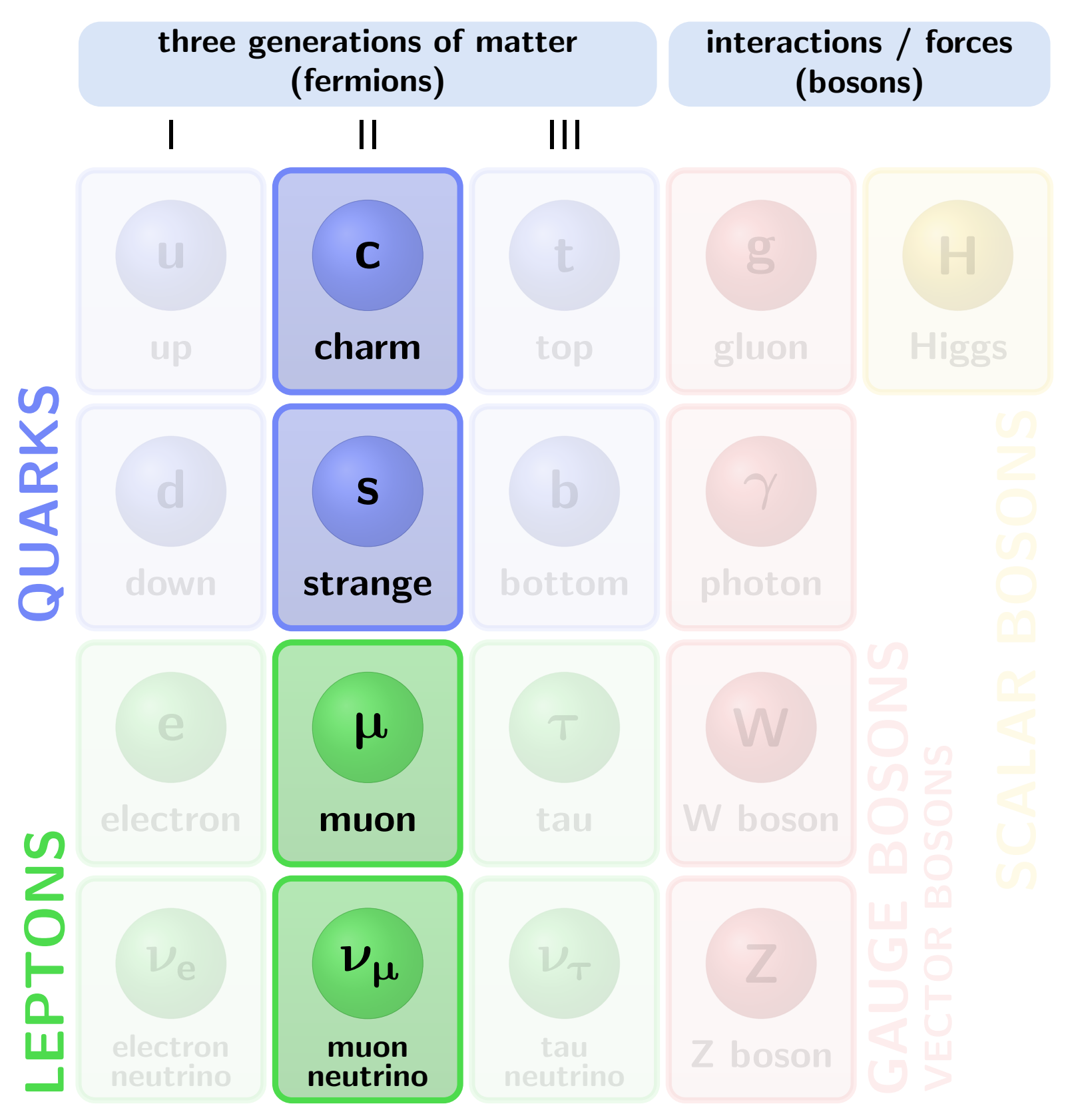

three generations of matter
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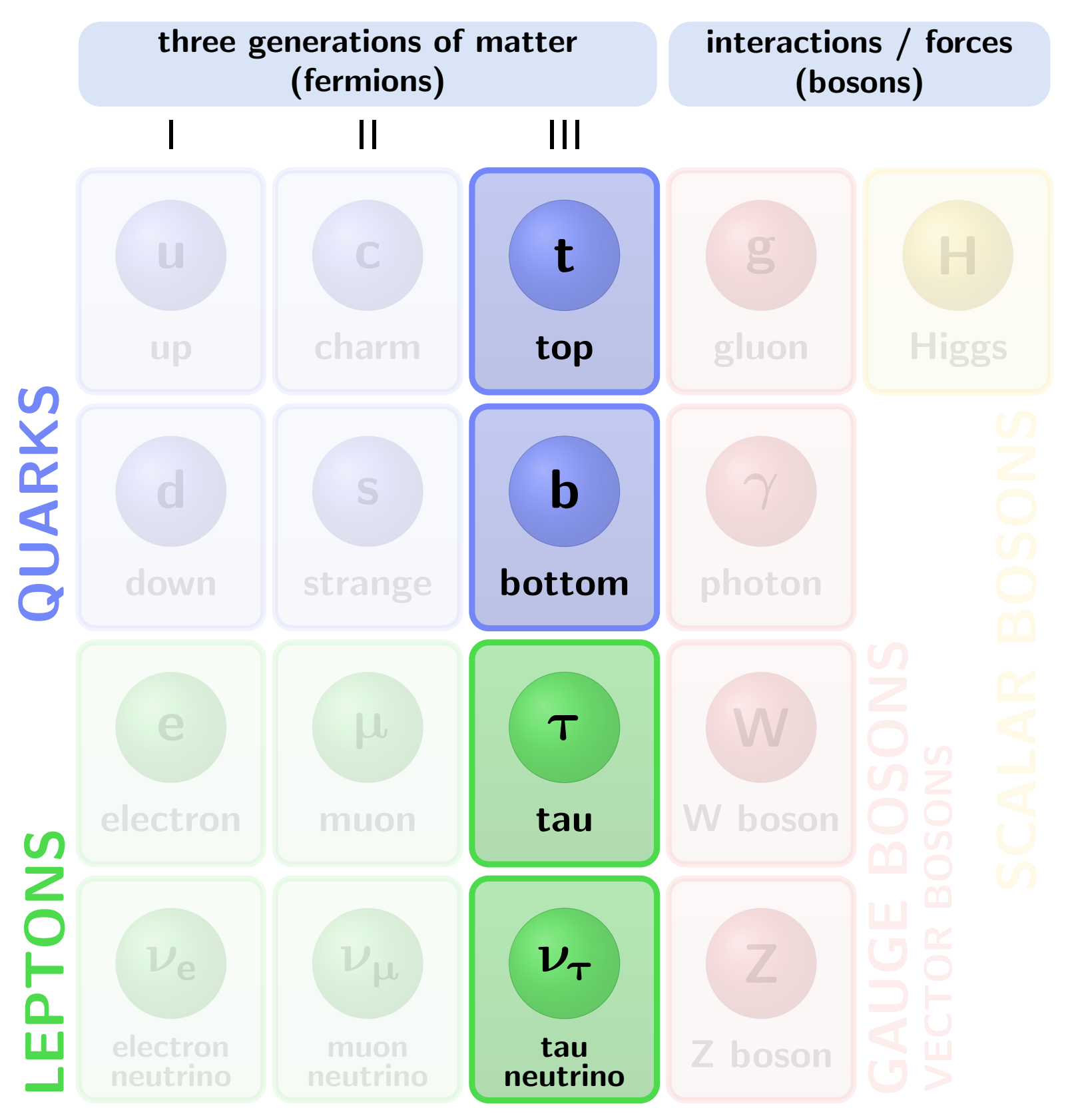

three generations of matter
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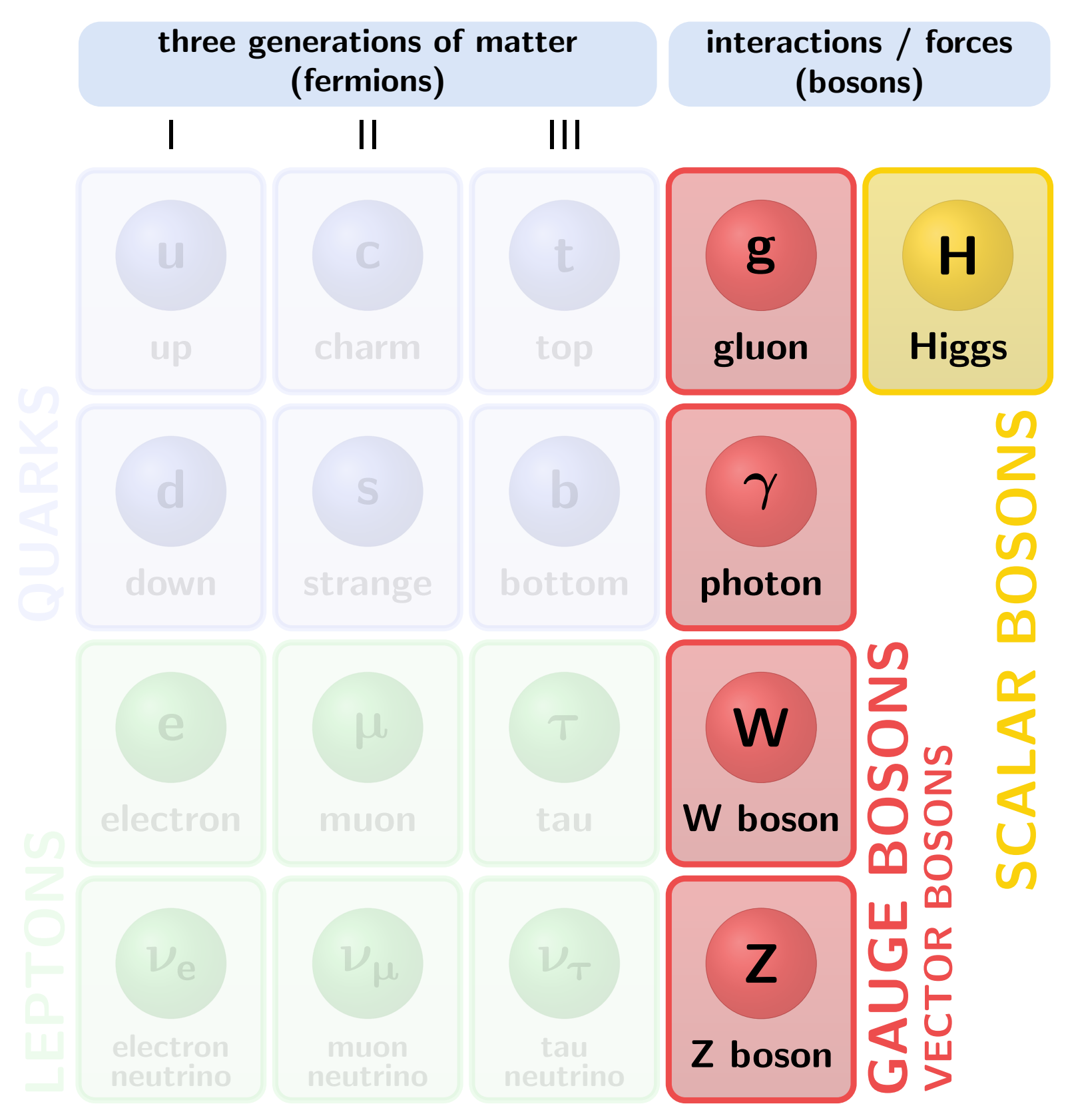

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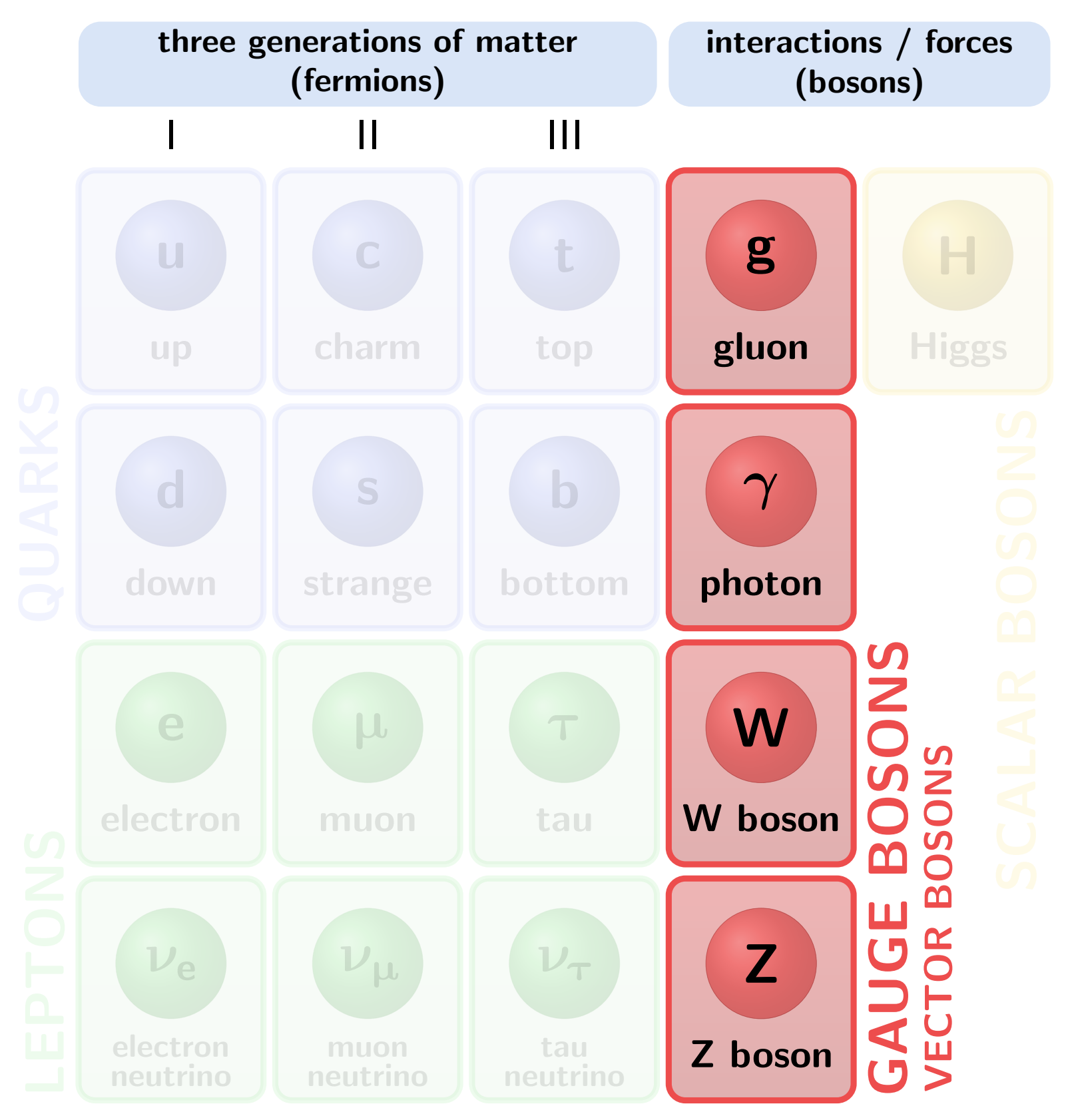

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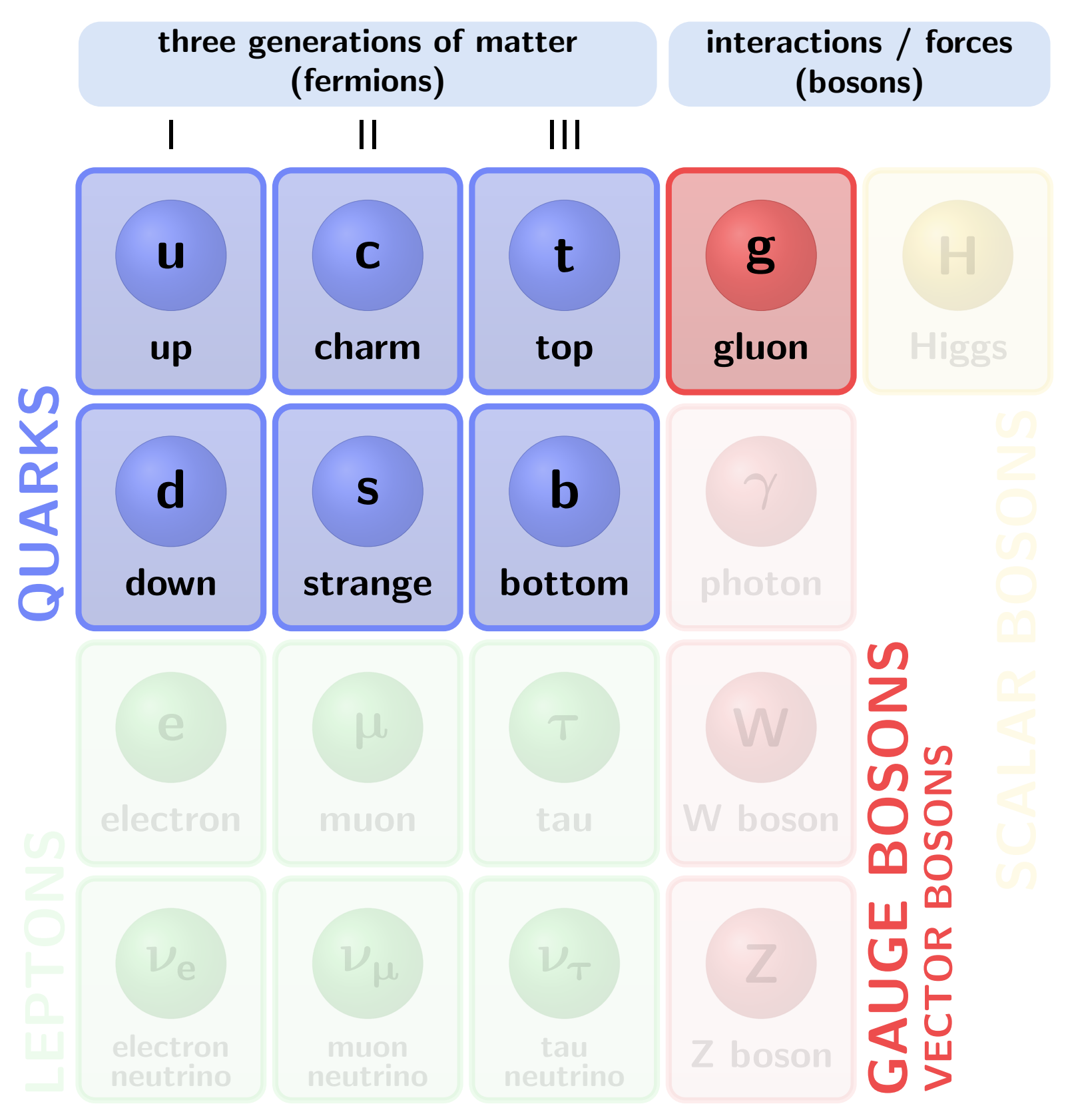

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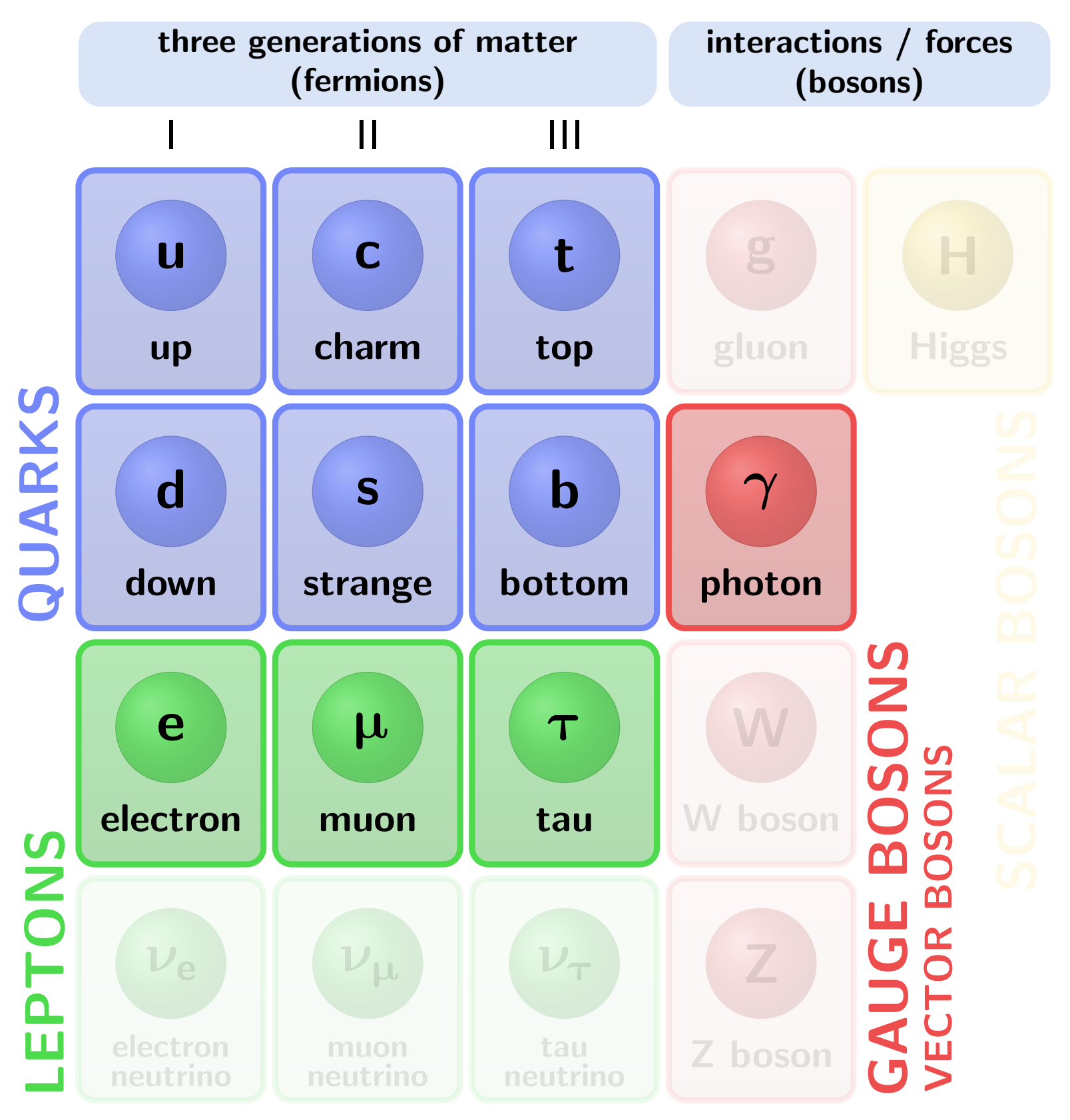

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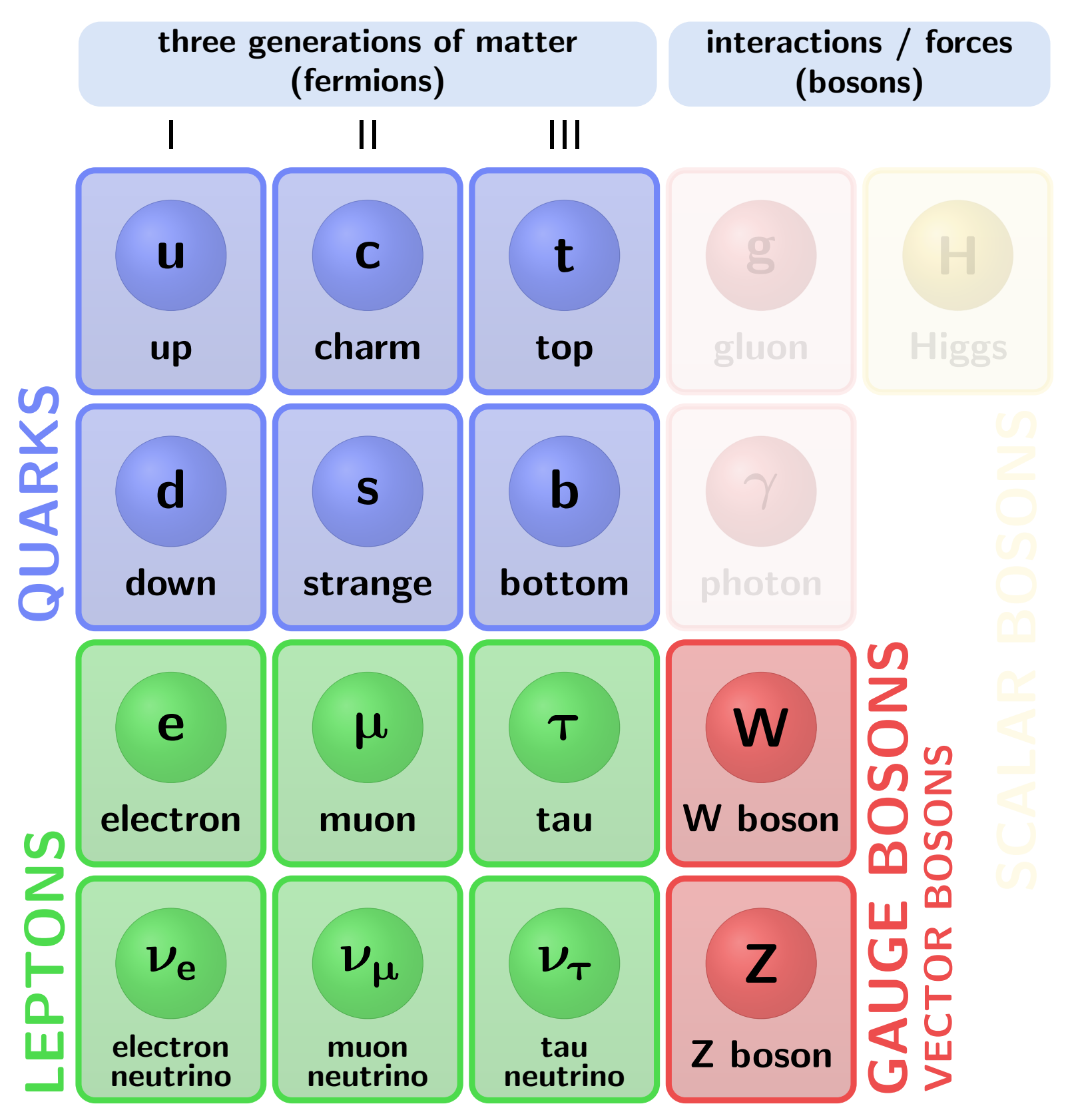

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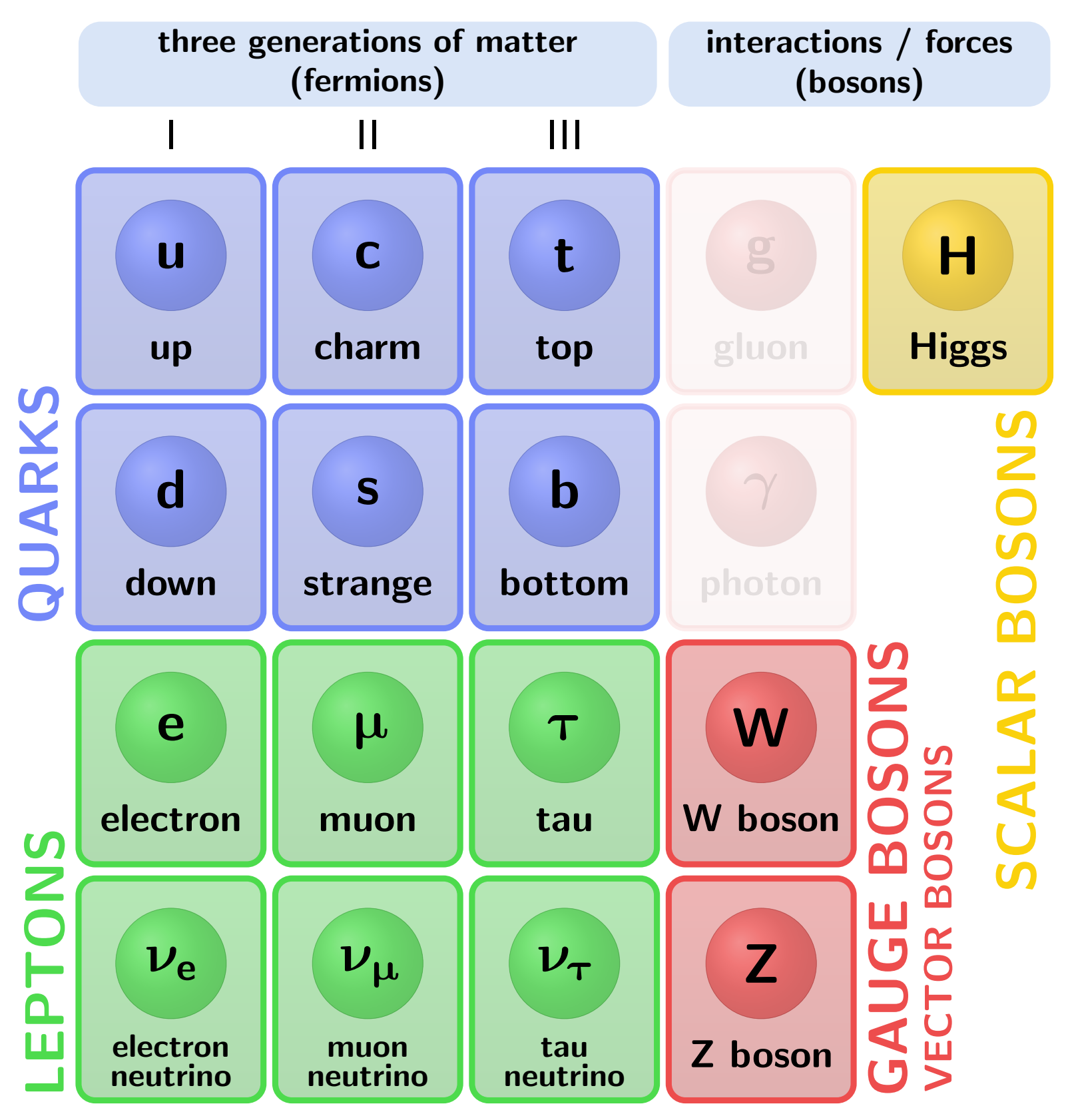

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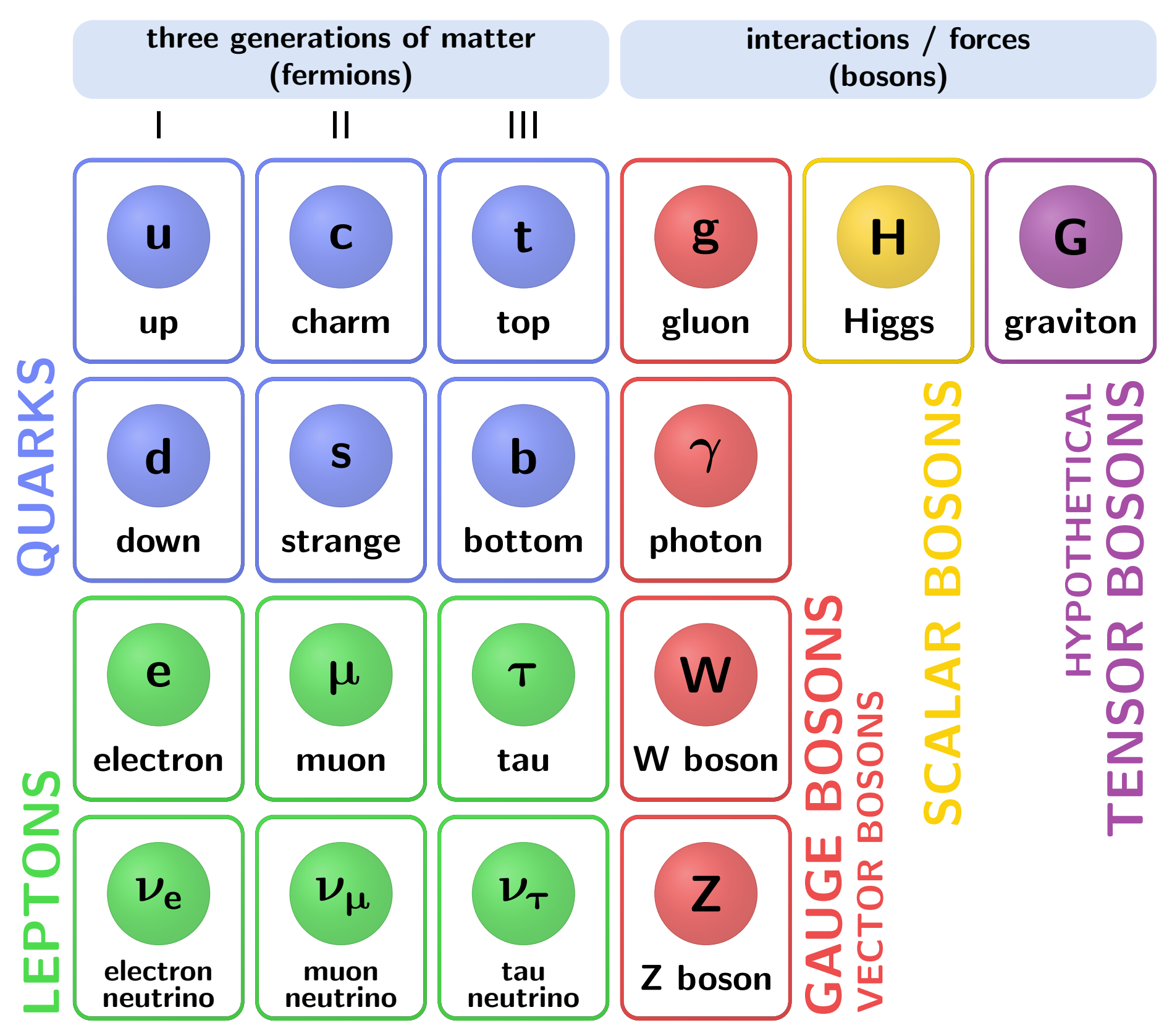

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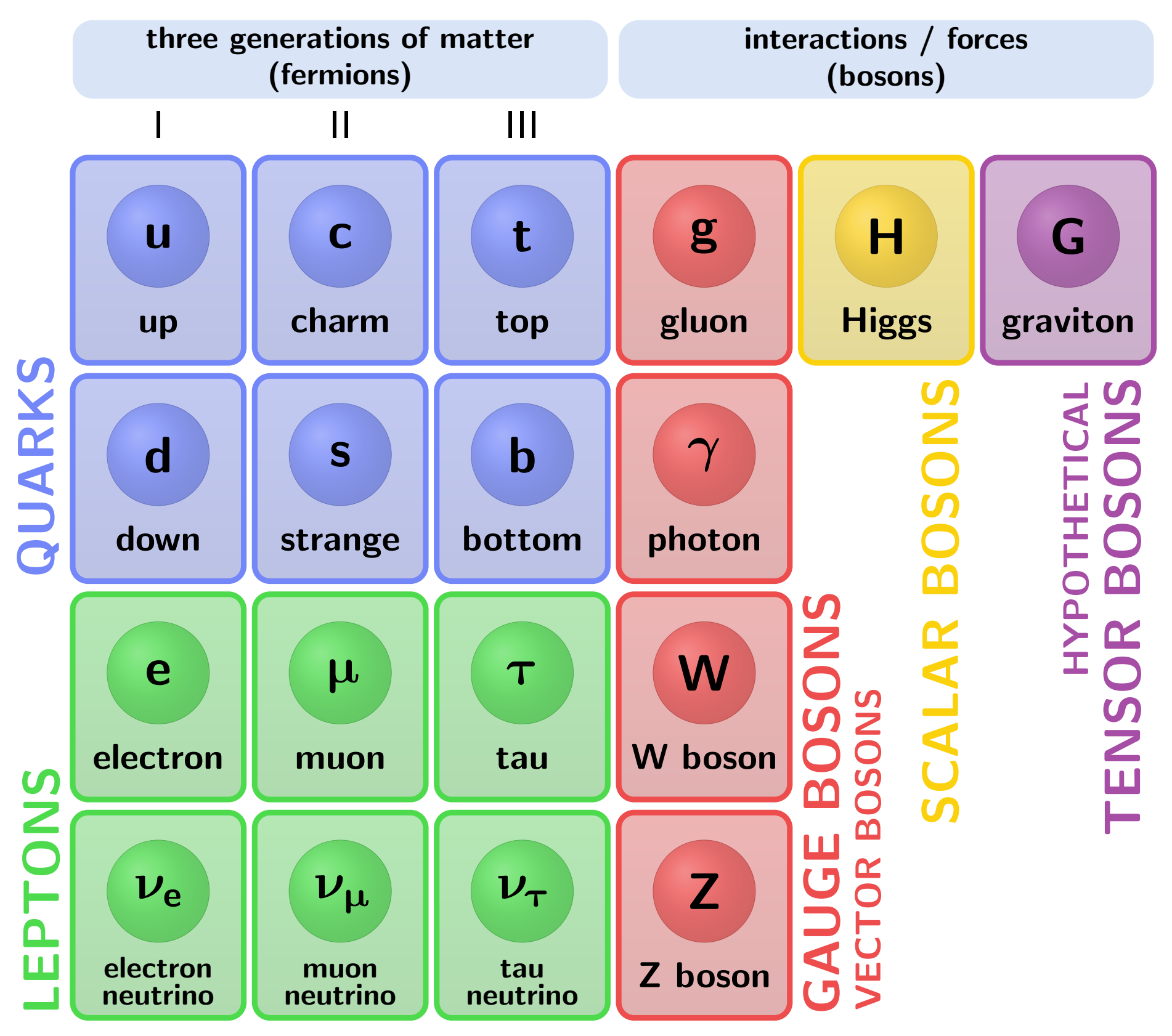

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VECTOR BOSONS
SCALAR BOSONS

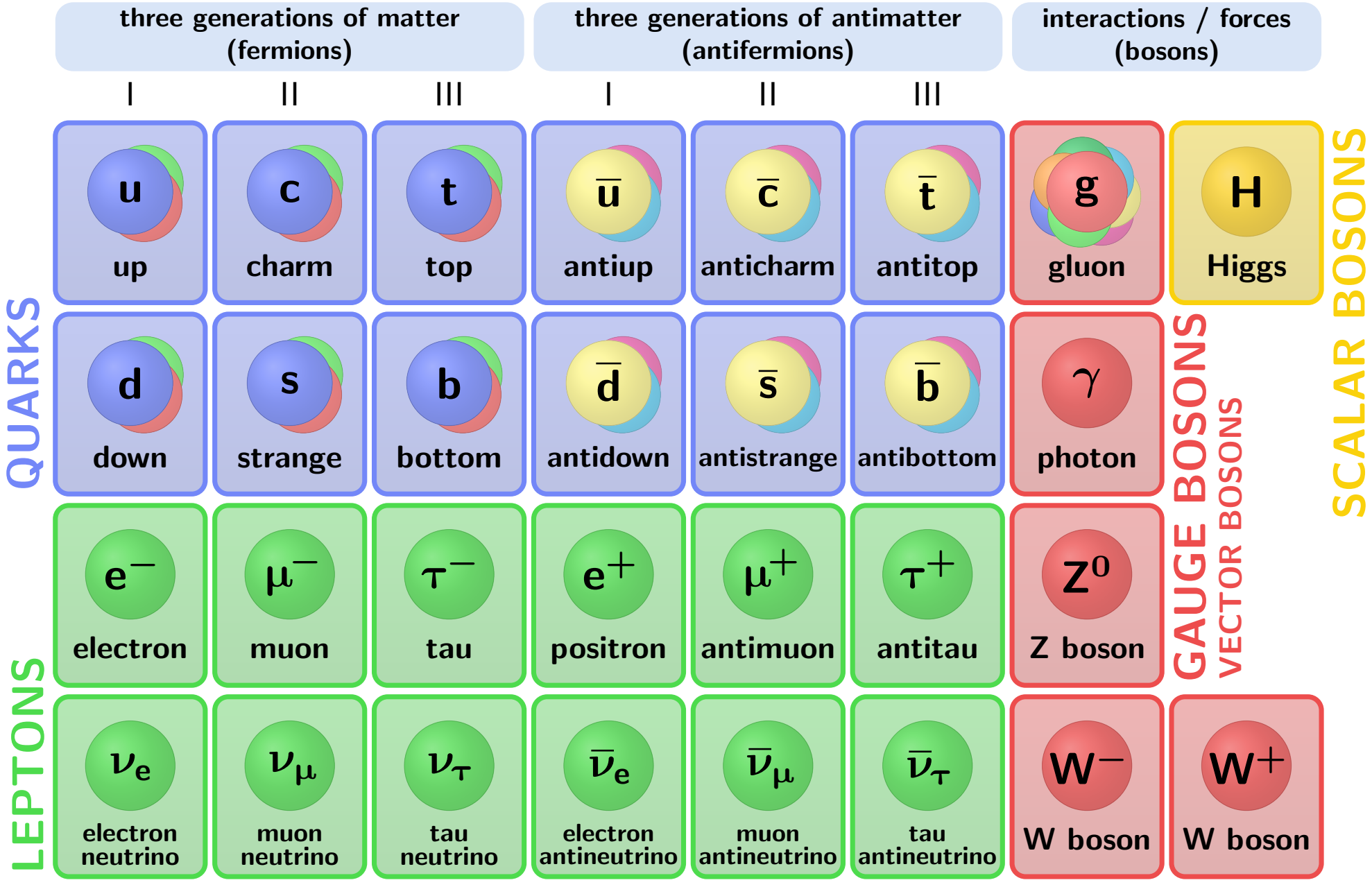

three generations of matter
(fermions)
interactions / forces
(bosons)
I
II
III
u
up
c
charm
t
top
g
gluon
H
Higgs
d
down
s
strange
b
bottom
γ
photon
e
electron
μ
muon
τ
tau
W
W boson
νe
electron neutrino
νμ
muon neutrino
ντ
tau neutrino
Z
Z boson
QUARKS
LEPTONS
GAUGE BOSONS
VECTOR BOSONS
SCALAR BOSONS

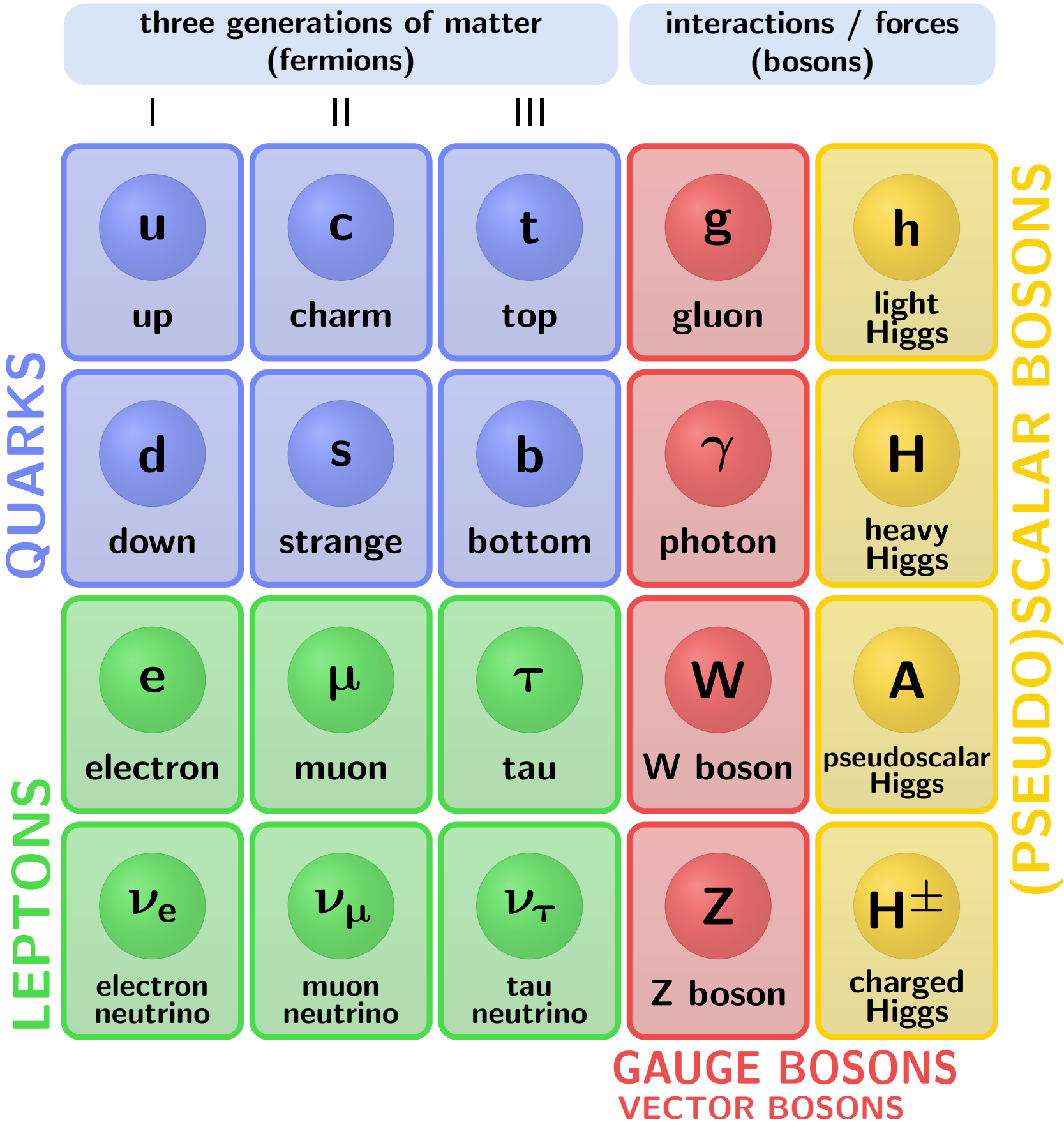

three generations of matter (fermions)
interactions / forces (bosons)
I
II
III
u
up
c
charm
t
top
g
gluon
H
Higgs
d
down
s
strange
b
bottom
γ
photon
QUARKS
e
electron
μ
muon
τ
tau
W
W boson
LEPTONS
νe
electron neutrino
νμ
muon neutrino
ντ
tau neutrino
Z
Z boson
GAUGE BOSONS
VECTOR BOSONS
SCALAR BOSONS

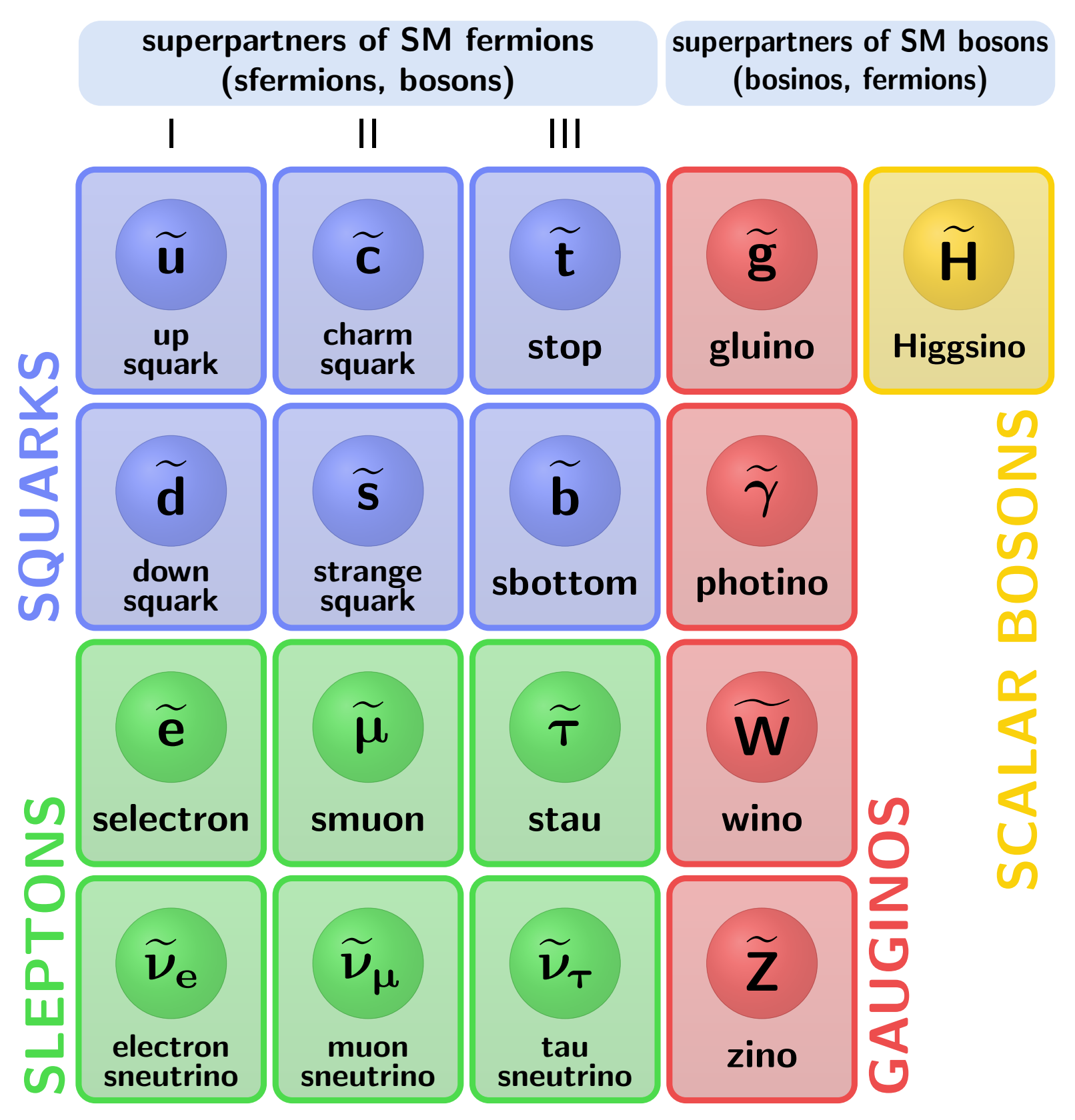

three generations of matter
(fermions)
interactions / forces
(bosons)
I
II
III
QUARKS
u
up
c
charm
t
top
g
gluon
H
Higgs
SCALAR BOSONS
d
down
s
strange
b
bottom
γ
photon
LEPTONS
e
electron
μ
muon
τ
tau
W
W boson
GAUGE BOSONS
VECTOR BOSONS
ν_e
electron neutrino
ν_μ
muon neutrino
ν_τ
tau neutrino
Z
Z boson

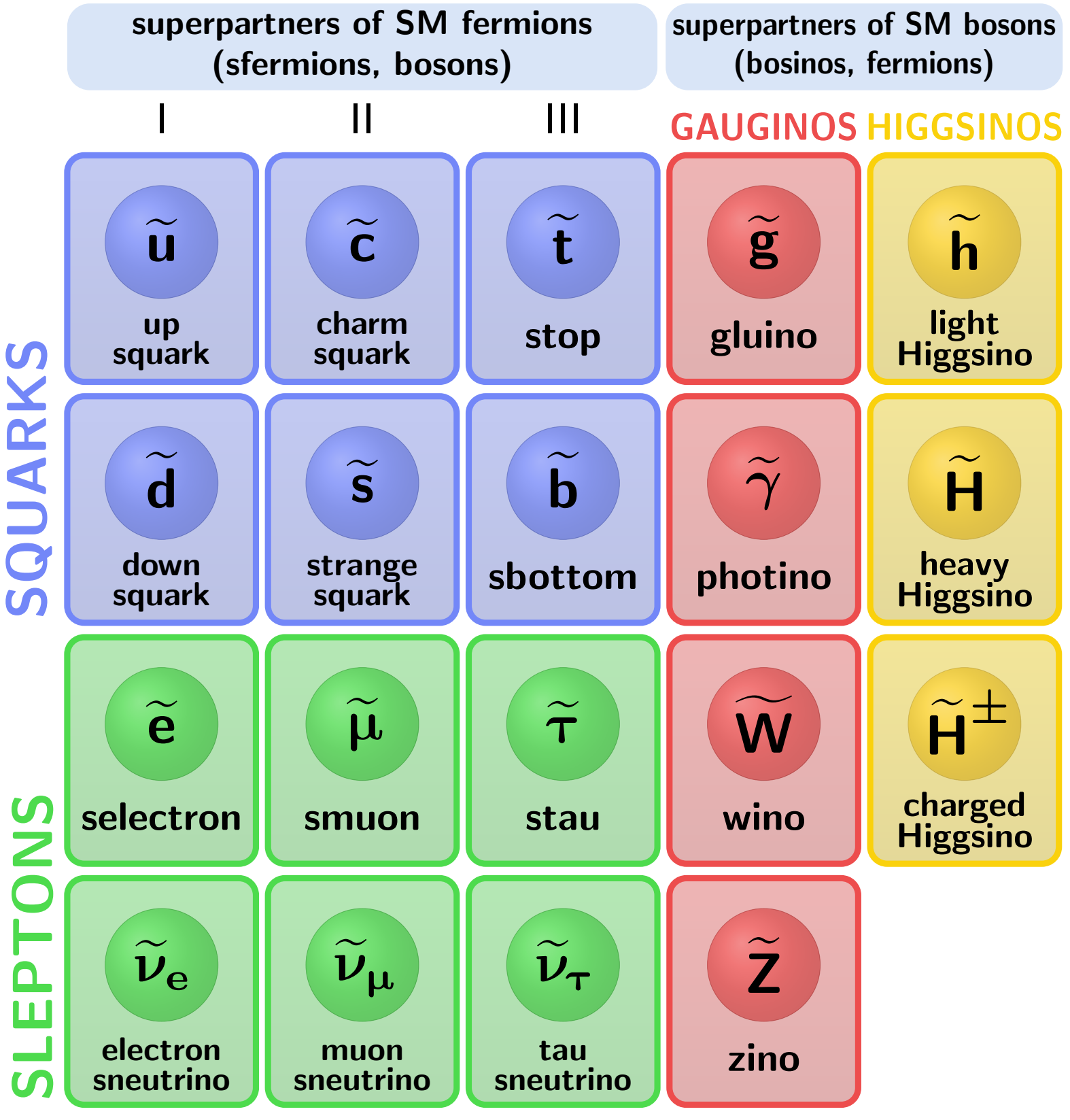

three generations of matter
(fermions)
interactions / forces
(bosons)
I
II
III
u
up
c
charm
t
top
g
gluon
H
Higgs
G
graviton
d
down
s
strange
b
bottom
γ
photon
e
electron
μ
muon
τ
tau
W
W boson
ν_e
electron neutrino
ν_μ
muon neutrino
ν_τ
tau neutrino
Z
Z boson
QUARKS
LEPTONS
GAUGE BOSONS
VECTOR BOSONS
SCALAR BOSONS
HYPOTHETICAL
TENSOR BOSONS

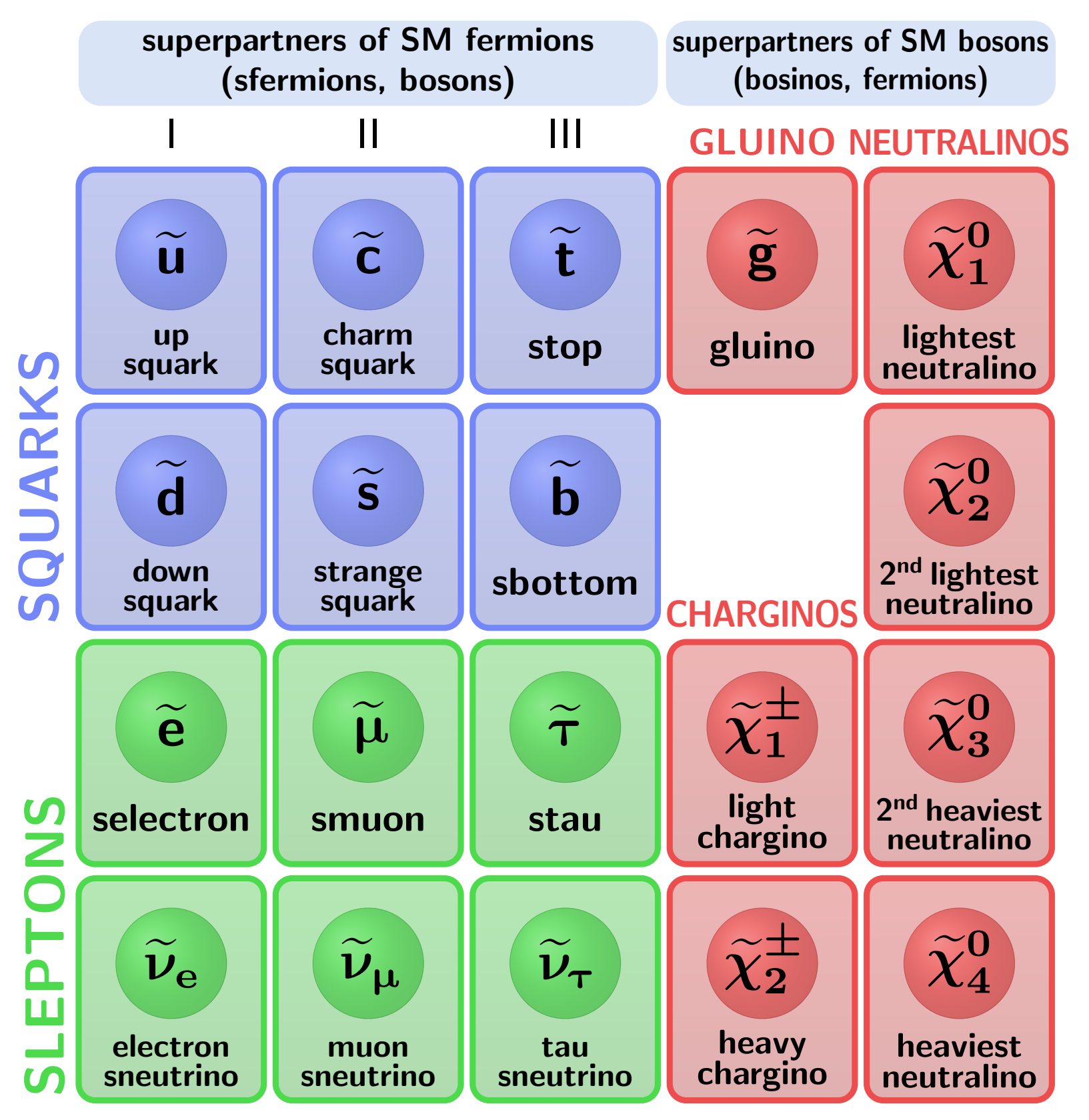

three generations of matter
(fermions)
interactions / forces
(bosons)
I
II
III
QUARKS
LEPTONS
GAUGE BOSONS
VECTOR BOSONS
SCALAR BOSONS
HYPOTHETICAL
TENSOR BOSONS
u
up
c
charm
t
top
g
gluon
H
Higgs
G
graviton
d
down
s
strange
b
bottom
γ
photon
e
electron
μ
muon
τ
tau
W
W boson
νe
electron neutrino
νμ
muon neutrino
ντ
tau neutrino
Z
Z boson


three generations of matter (fermions)
three generations of antimatter (antifermions)
interactions / forces (bosons)
I
II
III
I
II
III
QUARKS
LEPTONS
GAUGE BOSONS
VECTOR BOSONS
SCALAR BOSONS
u
up
c
charm
t
top
ū
antiup
c̄
anticharm
t̄
antitop
g
gluon
H
Higgs
d
down
s
strange
b
bottom
d̄
antidown
s̄
antistrange
b̄
antibottom
γ
photon
e⁻
electron
μ⁻
muon
τ⁻
tau
e⁺
positron
μ⁺
antimuon
τ⁺
antitau
Z⁰
Z boson
νe
electron neutrino
νμ
muon neutrino
ντ
tau neutrino
ν̄e
electron antineutrino
ν̄μ
muon antineutrino
ν̄τ
tau antineutrino
W⁻
W boson
W⁺
W boson


three generations of matter
(fermions)
interactions / forces
(bosons)
I
II
III
QUARKS
LEPTONS
(PSEUDO)SCALAR BOSONS
GAUGE BOSONS
VECTOR BOSONS
u
up
c
charm
t
top
g
gluon
h
light
Higgs
d
down
s
strange
b
bottom
γ
photon
H
heavy
Higgs
e
electron
μ
muon
τ
tau
W
W boson
A
pseudoscalar
Higgs
νe
electron
neutrino
νμ
muon
neutrino
ντ
tau
neutrino
Z
Z boson
H±
charged
Higgs


superpartners of SM fermions
(sfermions, bosons)
superpartners of SM bosons
(bosinos, fermions)
I
II
III
SQUARKS
SLEPTONS
GAUGINOS
SCALAR BOSONS
$\tilde{u}$
up squark
$\tilde{c}$
charm squark
$\tilde{t}$
stop
$\tilde{g}$
gluino
$\tilde{H}$
Higgsino
$\tilde{d}$
down squark
$\tilde{s}$
strange squark
$\tilde{b}$
sbottom
$\tilde{\gamma}$
photino
$\tilde{e}$
selectron
$\tilde{\mu}$
smuon
$\tilde{\tau}$
stau
$\tilde{W}$
wino
$\tilde{\nu}_e$
electron sneutrino
$\tilde{\nu}_\mu$
muon sneutrino
$\tilde{\nu}_\tau$
tau sneutrino
$\tilde{Z}$
zino


superpartners of SM fermions
(sfermions, bosons)
superpartners of SM bosons
(bosinos, fermions)
I
II
III
GAUGINOS
HIGGSINOS
SQUARKS
SLEPTONS
ũ
up squark
c̃
charm squark
t̃
stop
g̃
gluino
h̃
light Higgsino
d̃
down squark
s̃
strange squark
b̃
sbottom
γ̃
photino
H̃
heavy Higgsino
ẽ
selectron
μ̃
smuon
τ̃
stau
W̃
wino
H̃±
charged Higgsino
ν̃e
electron sneutrino
ν̃μ
muon sneutrino
ν̃τ
tau sneutrino
Z̃
zino


superpartners of SM fermions
(sfermions, bosons)
superpartners of SM bosons
(bosinos, fermions)
I
II
III
GLUINO NEUTRALINOS
SQUARKS
$\tilde{u}$
up squark
$\tilde{c}$
charm squark
$\tilde{t}$
stop
$\tilde{g}$
gluino
$\tilde{\chi}_1^0$
lightest neutralino
$\tilde{d}$
down squark
$\tilde{s}$
strange squark
$\tilde{b}$
sbottom
$\tilde{\chi}_2^0$
2nd lightest neutralino
CHARGINOS
SLEPTONS
$\tilde{e}$
selectron
$\tilde{\mu}$
smuon
$\tilde{\tau}$
stau
$\tilde{\chi}_1^\pm$
light chargino
$\tilde{\chi}_3^0$
2nd heaviest neutralino
$\tilde{\nu}_e$
electron sneutrino
$\tilde{\nu}_\mu$
muon sneutrino
$\tilde{\nu}_\tau$
tau sneutrino
$\tilde{\chi}_2^\pm$
heavy chargino
$\tilde{\chi}_4^0$
heaviest neutralino